

Influential Observations and Inference in Accounting Research

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Abstract

Accounting studies routinely encounter observations taking on extreme values. Such observations can influence statistical estimates (coefficient) and inferences. Our survey of the accounting literature documents that the two most common approaches used to address influential observations are winsorization and truncation. Although widely used, there is little systematic evidence in the literature as to their efficacy. While expedient, both approaches depend on researcher-selected cutoffs, are often applied on a variable-by-variable basis, and can alter legitimate data (i.e., valid) points. We compare winsorization and truncation with influence diagnostics (e.g. Cook's distance) and robust regression (RR), the latter two of which are suggested in the statistics and econometrics literatures. We replicate three published studies to show how the choice of method to account for influential observations impacts estimates and inferences. We also use simulations to compare the alternative approaches. We find that winsorization and truncation are largely ineffective in mitigating the effect of influential observations. Although both influence diagnostics and RR outperform winsorization and truncation, RR generally outperforms all other methods. RR, which focuses on model fit to deal with influential observations, is relatively straightforward to implement in accounting settings. While RR is not without limitations, overall, our findings lead us to recommend that future accounting studies consider using RR, or at least report sensitivity tests using RR.

1. Introduction

Archival accounting research studies routinely encounter observations (i.e., data points) taking on extreme values that can impact statistical estimates and inferences. Although observations with extreme values can occur in any research setting (e.g., experimental or quasi-experimental settings), the incidence of extreme data points is arguably more pervasive in empirical accounting studies. One reason is cross-sectional and inter-temporal variation in the type and nature of the economic activities that firms must report on (i.e., account for). Another reason is the nature of GAAP measurement rules which give rise to firm-level accounting numbers that are often based on estimates and assumptions. A final reason is the quasi-experimental nature of most empirical accounting research, which means that researchers must rely on observational data obtained from commercial databases as opposed to data generated and measured in a tightly controlled experimental setting.¹

Surprisingly, there is little systematic evidence in the accounting literature on the efficacy and trade-offs associated with alternative approaches to identify and mitigate the effect of influential observations. Our study fills this void by empirically documenting the shortcomings of the approaches commonly used in accounting (namely, winsorization and truncation) to deal with influential observations, and by documenting the properties and advantages of approaches recommended in the statistics and econometrics literatures (specifically, influence diagnostics and robust regression, hereafter, RR). Since accounting researchers typically use linear models (OLS and its variations) we use these models in our comparative analysis of the alternative approaches to identify and treat influential observations.^{2,3}

If an observation taking on an extreme value is simply a data error, it should be corrected (if possible) or eliminated from the sample if it cannot be corrected. However, extreme values will naturally arise in the production of accounting numbers due to unusual or infrequent economic events affecting a

¹ We use the terms archival accounting research studies, accounting studies, and accounting settings, interchangeably, to refer to studies examining samples drawn from databases such as COMPUSTAT, CRPS, I/B/E/S, Thomson Reuters Worldscope, Audit Analytics, etc. A large number of these studies examine questions related to the production and use of accounting information by capital market participants.

² In addition to influential observations, we recognize that other sources of model misspecification are also important (e.g., nonlinearity). While of interest in their own right, examination of such issues is beyond the scope of our study.

³ Although our study is the first to provide a comprehensive analysis of the effect of influential observations by comparing alternative approaches to identify and address them in accounting settings, we do note that this issue has not gone unnoticed in accounting (see, e.g., Kothari et al. 2005; Kraft, et al 2006; and, Teoh and Zhang 2011).

subset firms in the overall sample. In accounting settings, there is no shortage of real economic events that will produce observations taking on extreme values, including, but are not limited to, corporate reorganizations, stock-offerings, one-time gains/losses, acquisitions, divestitures, restructurings, impairments, big baths, etc. For example, a firm eliminating a business unit (an infrequent economic event) would likely record a large restructuring and/or impairment charge producing very low total accruals and earnings due to that economic event, which in turn might cause the firm to have a large positive or negative stock return. The key point is that the extreme values in these cases are legitimate observations driven by real economic events (i.e., they are not data errors), which arguably should be retained in the sample.⁴

Although a precise definition is elusive, we use the term *influential observations* to describe data values that are likely to affect statistical estimates (coefficients) and inferences from the linear estimation methods (such as OLS regression) commonly used in accounting. It is important to stress at the outset that *outliers* identified by evaluating the distribution of each variable (as is done under truncation and winsorization) are not necessarily influential observations in a regression context. For example, if an extreme value of the dependent variable y is paired with an extreme value of the independent variable x , and the data point (y,x) falls in line with the other data points, then that (y,x) data point will not necessarily be an *influential observation* in a regression context. Consistent with this, statistics and econometrics textbooks do not typically recommend winsorization or truncation, that is, they do not recommend methods that modify or delete data prior to estimating a regression model.

Our survey of the accounting literature finds that most studies attempt to (ex-ante) mitigate the impact of influential observations by using winsorization or truncation. Under these approaches, before conducting any statistical tests, a researcher alters the original data, likely affecting legitimate observations in

⁴ To provide some concrete examples: using all non-financial firms on COMPUSTAT with data available for 2014 we identified 4,458 observations to analyze [we excluded firms without a CIK number or SIC code or the data to calculate the market value of equity; return on assets ($ROA = \text{Net Income} / \text{Total Assets}$); and accruals as percentage of total assets ($ACC = (\text{Net Income} - \text{Net Cash Flow from Operating Activities}) / \text{Total Assets}$)]. In this sample, the 99th percentile of ROA (ACC) is 0.38 (0.59). In the ROA distribution, one firm, Terra Nitrogen (CIK 0000879575) has an ROA of 0.89 (1.43 in 2013). Of significance is that, in both years, the “extreme” ROAs are driven by the economics of Terra’s business (its high margins). The key point is that these ROAs are legitimate data values, yet such observations would invariably be altered by the standard winsorization and truncation approaches used in accounting. As an example using accruals, Vanda Pharmaceuticals (CIK 0001347178) had ACC of 0.69 in 2014, which is driven by actual economic events (Vanda reached a legal settlement, accruing revenue of \$77.6 million in the process). Again, the point is that this observation is a legitimate data value, yet its value would invariably be altered by the winsorization and truncation approaches used in accounting.

the process, by truncating or winsorizing extreme values of each variable using cutoffs like the 1st and 99th percentiles of each variable's distribution. Other important aspects of both approaches are that they are *ex ante* in nature; they depend on researcher-selected cutoffs; and are often applied on a variable-by-variable basis. While these approaches are the most widely used in accounting, likely due to their expediency, we find little evidence to indicate that the decision to adopt either approach is systematically tailored to or driven by special characteristics of the data or by the nature of the theory being tested. Another important aspect of our survey's results is that we find considerable variation in the way researchers implement winsorization and truncation. Such variation has the distinct disadvantage of reducing the inter-study comparability of results, as well as inducing sample-specific and/or time-period specific inferences.

In the typical accounting setting, where researchers generally implement linear models (OLS and its variations), ignoring the underlying cause(s) of observations with extreme values is problematic because such observations are likely to affect OLS coefficient estimates and inferences. This is because OLS is known to be very sensitive to extreme observations, both in the dependent and independent variables. While OLS yields unbiased estimates, even in the presence of extreme observations or skewness in the distributions of the variables of interest, such estimates reflect the "average" effect, which is typically not the most appropriate statistic for hypothesis testing for skewed distributions. Although it is theoretically possible to modify an OLS model to include additional variables or nonlinear terms designed to capture the effect of unusual, but actual economic events (i.e., in attempt to control for such events), it is not always easy for a researcher to identify the sources of these economic events (or to measure them). On the other hand, research in statistics and econometrics has developed a number of influence diagnostics and robust estimation methods which are designed to address influential observations. Influence diagnostics generally rely on the distribution of OLS errors and the changes in estimates when a single observation or a few observations are deleted from the sample (e.g., Cook's distance and DFFITS, see Belsley et al. 1980). Robust estimation methods can be viewed as a generalization of influence diagnostics and include a family of estimation methods generally known as robust regression (RR). Conceptually, RR provides a statistical compromise by using all of the original (raw) data points, but down-weighting influential observations, versus the *ex-ante* exclusion or alteration, usually arbitrarily, of some of the original data (which characterize winsorization and

truncation). A potential compelling rationale to consider using RR in accounting settings is that the RR literature has analytically and empirically demonstrated the validity of RR estimators (see Andersen 2008 for a literature review). Among the available RR estimation methods, we focus on MM-estimation (see section 4). While not often used in accounting, RR is a viable alternative to winsorization, truncation, and influence diagnostics when it comes to identifying and mitigating the effect of influential observations.

In this study, we evaluate the effectiveness of the alternative methods discussed above by first replicating three published accounting studies to show how winsorization, truncation, influence diagnostics, and RR affect coefficient estimates and inferences. The objective of the replications is not to overturn prior studies, but rather to demonstrate how the common practices used in accounting can yield results whose coefficient estimates are sensitive to influential observations, and thus, may not provide robust inferences. We next use simulations to document and compare the benefits, consequences, and trade-offs associated with each method. In a simulated (controlled) setting we can hold the data-generating process constant which enables us to more clearly compare the relative effectiveness of each approach. Our results indicate that winsorization and truncation are largely ineffective in dealing with truly influential observations. In particular, winsorizing does little to mitigate the effect of influential observations, while truncation imparts a bias towards zero in OLS coefficients. Although influence diagnostics out-perform winsorization and truncation, they are sensitive to the effects of “masking” and “swamping.” “Masking” and “swamping” occur when extreme observations are clustered in the data, and researchers erroneously identify too few of them. We find that RR (using MM-estimation) outperforms the other approaches, especially in the presence of observations driven by unusual economic events that are correlated with the dependent and independent variables. Our findings should enable future researchers to make more informed research design choices, as well as to better appreciate the consequences of choosing one alternative approach over another.⁵

Our findings lead us to recommend that future accounting studies consider using RR, or at a least report sensitivity tests based on RR because RR can be used in any setting where researchers use OLS. If, in fact there are no influential observations, RR and OLS will produce very similar coefficients and inferences.

⁵ Albeit in a different setting, like Gow, et al (2010); Larcker and Rusticus (2010); Lennox, et al (2012); and, Shipman, et al (2016), we draw on results from the statistics and econometrics literatures along with simulation analysis and replications of prior studies to provide researchers with systematic evidence to guide their future research design choices and to better appreciate the consequences (i.e., impact on inferences) of choosing one alternative over another.

An added benefit of RR is that influential observations can be quickly identified using a data-driven approach, perhaps leading to a revision in the theory being tested or to models that better explain the overall data. Nevertheless, we stress that there is no panacea when it comes to identifying and treating influential observations because all approaches invariably require judgment on researchers' part. For example, RR requires careful examination of the data and familiarity with RR estimation techniques. Moreover, while it is important for us to address the effect of influential observations on inferences, other sources of model misspecification should not be ignored and are worth investigating as well.

The paper is organized as follows. Section 2 reports survey evidence documenting the approaches used to account for influential observations in accounting, along with a brief discussion of the approaches suggested in statistics and econometrics. Section 3 describes the consequences of influential observations in the OLS setting. Section 4 describes RR estimation. To assess the effect of alternative approaches to treat influential observations in actual accounting settings, section 5 reports the results of replicating three studies. Section 6 describes the simulation analyses and reports the results. Section 7 concludes.

2. Approaches used to Identify and Treat Influential Observations in Accounting Research Studies

2.1 Survey findings

To document how accounting researchers identify and treat influential observations, we reviewed 1,833 studies published from 2006 to 2014 in *The Accounting Review*, *Journal of Accounting Research*, *Journal of Accounting and Economics*, *Review of Accounting Studies*, and *Contemporary Accounting Research*. Surveyed studies include, but are not limited to, auditing, analysts' forecasts, management compensation, earnings management, conservatism, taxes, disclosure, and the earnings-returns relation.⁶

As shown in Panel A of Table 1, of the 1,833 studies reviewed, 68% (1,252) are archival in nature, and of those 1,252, only 68% (N= 851) mention influential observations or describe their approach to identify and treat such observations. In other words, nearly one-third (32%) do not mention influential observations at all, or do not clearly describe their approach to identify such observations. Turning to the 851 studies mentioning influential observations, 63% (N=532) use winsorization; 30% (N=258) use truncation, 7% (N=59)

⁶ Our survey may have limitations. For example, it cannot reflect cases (however unlikely) where a researcher took steps to address influential observations, but did not mention that in the paper.

use both winsorization and truncation; 11% (N=94) use variable ranks (rather than continuous values of the raw data); and 20% (N=173) use a variety of techniques including various forms of residual analysis and RR (N and % do not sum to 851 and 100% because some studies report using multiple approaches).⁷ In summary, the most common approaches to address influential observations in archival studies are winsorization and truncation, applied to both the dependent and independent variables.

Before turning to Panels B and C and the details of how winsorization and truncation are implemented, some other aspects of the archival studies summarized in Panel A deserve mention. For example, in those 1,252 studies, stock returns are the dependent variable 16% (N=202) of the time, and in the majority of those cases researchers alter the return distribution. Specifically, 23% (N=46) of the studies winsorize returns when it is the dependent variable; 23% (N=46) truncate the return distribution; 9% (N=18) use the rank of returns; 14% (N=29) report using other techniques; while 26% (N=52) do not mention the presence of extreme data points (see the row labeled “Influential/extreme observations not mentioned” at the bottom of Panel A).

As reported in Panel A, 63% of the archival studies (N=532 of 851) reported winsorizing at least one variable, making it the most common approach used in accounting. To provide texture on how winsorization is applied, Panel B breaks down the studies using winsorization. Of those 532 studies, 63% (N=337) winsorize all variables; 6% (N=32) winsorize only the dependent variable; 10% (N=51) report winsorizing all of the independent variables; 18% (N=95) “Leave out some variables;” and in 3% (N=17) of the studies the researcher’s approach is unclearly stated (see “Unclear cases” in Panel B). “Leave out some variables” are those studies that did not apply their stated winsorization rule to all of their variables. “Unclear cases” are those studies where the text was not clear enough to determine which variables were winsorized and which weren’t (e.g., a study states “we winsorize extreme values” without stating which variables). While untabulated, some studies report using one winsorization rule for most variables, but then use an *ad-hoc* rule for other variables. Examples of *ad-hoc* rules are setting values of discretionary accruals (as a percent of total assets) that are greater (less) than 200% (-200%) to ± 2 ; winsorizing all variables except the log of total assets; and setting values of effective tax rates that are greater than one (or negative) to one (zero).

⁷ An example of a study using RR is Huson et al. (2012, p. 250) who state: “Our results on negative discretionary accruals are sensitive to the method of dealing with outliers (trimming versus robust regressions). The difference between the results from the trimmed and robust regressions is attributable to the trimmed accounting variables.”

To provide texture on how truncation is applied, Panel C provides a breakdown of the studies using truncation. Under truncation, ex-ante, the researcher imposes an upper and lower bound on each variable's distribution, discarding observations beyond those cutoffs (cutoffs are typically percentiles of a variable's distribution). As shown (in Panel A), truncation is the second most common procedure with 30% (N=258 out of 851) of the archival studies truncating at least one variable. Panel C reveals that of those 258 studies, 31% (N=79) truncate all variables; 9% (N=22) truncate only the dependent variable; 4% (N=10) report truncating all of the independent variables; 51% (N=132) "Leave out some variables;" and in 6% (N=15) of the studies the researcher's approach is unclearly stated (see "Unclear cases" in Panel C).⁸ While un-tabulated, when choosing truncation rules, many studies truncate observations using non-percentile cutoffs such as truncation of earnings scaled by total assets at ± 3 and truncation of firms with stock prices $< \$5$.

Having documented the approaches used across a variety of studies, Panel D reports on a specific research area, namely "Conservatism." Conservatism studies examine issues such as the benefits of conservatism (often referred to as timely loss recognition). We separately summarize conservatism studies because it is an active area and because the distributions of the variables of interest tend to be characterized by extreme values. The presence of extreme values raises the concern that those data points have the potential to be influential observations in the standard regression models used in conservatism studies. Panel D's findings reveal that the approach to address influential observations varies widely across studies. For example, of the 57 studies identified, 37% (N=21) use winsorization, 35% (N=20) use truncation, 5% (N=3) report using both winsorization and truncation, 16% (N=9) use variable ranks, 14% (N=8) report using other techniques, and 30% (N=17) make no mention of how potential influential observations were addressed, if at all (N and % do not sum to 57 and 100% because some studies use multiple approaches). As in the overall sample, the most common approaches are winsorization and truncation. Given this similarity, one takeaway from Panel D that cannot be over-emphasized is that the approach used to deal with influential observations does not appear to be driven by the nature of the research question being addressed (e.g., conservatism) or by characteristics unique to the underlying data in a given area.

⁸ Analogous to the discussion of winsorization, "Leave out some variables" consists of studies that didn't apply their stated truncation rule to all of their variables while "Unclear cases" consists of studies where the text is not clear enough to determine which variables were truncated and which were not (e.g., a study states "we truncate extreme values" without stating which variables that applied to).

In summary, our survey evidence indicates that the procedures used to identify and treat influential observations in accounting are inconsistently applied across studies. Additional (un-tabulated) evidence supports that conclusion. First, in studies using winsorization and/or truncation, we identified numerous cases where a researcher used an *ad-hoc* rule for a subset of variables, while in a number of other studies the procedures used were unclearly stated (hence impossible to replicate). Second, while a number of studies report using the Belsley et al. (1980) regression diagnostics, most studies did not report enough detail as to the exact procedures/thresholds employed. Such inconsistencies make it difficult, if not impossible, to replicate those studies, to make comparisons with the results of related studies, or to reconcile conflicting results between studies. Finally, we found that discussion (even disclosure) of the methods used to deal with influential observations was very inconsistent across studies. Many studies relegated this important research design choice to footnotes in the text or in tables, while in numerous other studies such information was scattered throughout the text and/or across multiple footnotes.

2.2 Criteria for identifying influential observations in the econometrics and statistics literatures

To contrast accounting researchers' use of winsorization and truncation with the views and approaches currently held by statisticians and econometricians, in this section we briefly summarize the recommendations and "traditional" criteria for identifying influential observations that are found in statistics and econometrics textbooks. To be brief, we do not review these literatures in detail, but rather draw from the statistics (e.g., Belsley et al. 1980) and econometrics textbooks commonly used in PhD programs (e.g., Greene 2006; Wooldridge 2006; Angrist and Pischke 2009; Kennedy 2003). Our reading of these textbooks leads us to make two important observations. First, and perhaps most importantly, none mentions variable-by-variable winsorization or truncation as a recommended approach, even though these are the most common approaches in accounting. Second, methods for identifying and treating influential observations are generally mentioned only briefly in standard econometrics texts.

In the statistics literature, perhaps the most detailed discussion is found in Belsley et al. (1980) which discusses criteria for influential observations and explains the mechanics of several influence diagnostic tests. For example, Belsley et al. (1980, p. 3) state that "unusual or influential data points, of course, are not necessarily bad data points; they may contain some of the most interesting sample information." Belsley et

al. (1980) go on to state (pgs. 3-4) that “only after such data points have been identified can their quality be assessed and appropriate action taken. Such an analysis must invariably produce regression results in which the investigator has increased confidence. Indeed, this will be the case even if it is determined that no corrective action is required, for then the investigator will at least know that the data showing the greatest influence are legitimate.” Belsley et al. (1980) advocate diagnostic tests that generally rely on the distribution of OLS errors and the changes in estimates (i.e., leverage) when a single or few observations are deleted from the sample. Suggested diagnostics include: RSTUDENT (based on a cutoff of the distribution of standardized residuals); DFFITS (based on how much the regression function changes at the i th observation when the i th observation is deleted from the sample, which is similar to Cook’s distance, see Cook 1977); and DFBETAS (based on how much the coefficients change when the i th observation is deleted from the sample). All rely on examining the OLS residuals and the determination of influential observations using a cutoff criterion for “large” influence (recommended cutoffs to capture influential observations are $RSTUDENT > 2$, $DFFITS > 2/\sqrt{(p/n)}$; Cook’s distance $> 4/n$; and $DFBETA > 2/\sqrt{n}$, where n is the number of observations and p is the number of parameters). Such observations are then excluded from second- (or later) pass regressions estimated to draw a researcher’s “final” inferences.⁹

Turning to econometrics texts, we note the following discussion of influential observations. Section 4.9.3 in Greene (2006) devotes two pages to the issue, recommending the examination of OLS standardized residuals, referring readers to Belsley et al. (1980) for recommendations on cutoff values for standardized residuals. Section 9.4 in Wooldridge (2006) devotes a little less than five pages to the issue and notes that “sometimes, outliers are defined by the size of the residuals in an OLS regression where all observations are used. Generally, this is not a good idea.” Wooldridge (2006) also refers readers to Belsley et al. (1980) for additional guidance by stating “it is sometimes useful to have statistics that can detect such influential observations. These statistics do exist, but they are beyond the scope of this text (See, for example, Belsley et al. 1980).” Of note is that, Wooldridge (2006) advises that “OLS results should probably be reported with and without outlying observations in cases where one or several data points substantially change the results.”

⁹ These diagnostics are easily performed in SAS and Stata. Later in the paper when we perform our replications and simulations we use Cook’s distance along with winsorization, truncation, and RR.

Like these two texts, Kennedy (2003) refers readers to Belsley et al. (1980). In addition, chapter 20 of Kennedy (2003) briefly discusses influential observations and robust estimation, mentioning several robust estimators including: M-estimators, LP-estimators, L-estimators, trimmed least squares, and bounded estimators (weighted least squares). We find no mention of influential observations in Angrist and Pischke (2009). In summary, the most important takeaway from our brief discussion of how influential observations are discussed in econometrics and statistics textbooks is that none mention winsorization or truncation.

3. Influential Observations and OLS Estimation

In archival studies, OLS (or variations of it) is the most widely used methodology to estimate the relation between a dependent variable (y) and one or more independent variables (X). Under OLS we have:

$$y = \alpha + X\beta + \epsilon. \quad (1)$$

In (1), the estimated β is based on minimizing the sum of the squared errors. However, it can be shown that β may not necessarily reflect the “typical” effect of x on y for the bulk of the data because OLS estimates are based on the conditional mean of the dependent variable which means they suffer from the same problems as the mean itself, namely, they are susceptible to the effects of observations with influential values.

In theory, to address the effect of influential observations on the estimated parameters of an OLS model, a researcher could try to uncover what is causing the extreme values of x or y . Along these lines, the distribution of y will depend on the researchers’ sample selection procedure, the distribution of the x values, and unknown sources of variation captured by the error term. The key point here is simply that extreme values of y can be caused by a number of factors including: 1) extreme values of x that influence y ; 2) nonlinearities in the relation between x and y ; or, 3) additional variables that influence y , that are unknown to the researcher, or otherwise omitted from the model. As discussed below, it is important to recognize that not all extreme observations necessarily have the same effect on estimated OLS parameters. Simply put, not all extreme observations turn out to be influential observations. This highlights the conceptual weakness of winsorization and truncation, namely that both are ex-ante, ad-hoc and univariate nature which means that their arbitrary cutoff values are likely to alter legitimate observations (i.e., valid data points).

Observations taking on extreme values in either the dependent or independent variables can be the result of data errors or reflect actual economic events or real characteristics of the firms in the sample. For

example, observations taking on extreme values for the dependent variable can arise from skewness in the independent variables or from differences in the data generating process for a subset of the sample (e.g., the economic determinants of the return on equity of a firm in the start-up phase are not the same as those for a firm in the mature or decline phase). More generally, it is well known that the distributions of variables used in accounting research are often non-normal and exhibit considerable skewness. While OLS yields unbiased parameter estimates even in the presence of variable skewness, such (unbiased) estimates reflect the “average” effect, which is not typically the most appropriate statistic for hypothesis testing when the underlying distribution is skewed. For example, suppose a researcher is interested in testing the hypothesis that investors over-estimate the persistence of an accounting variable, which in turn leads to predictable returns. Here an OLS regression of future returns on that accounting variable would yield an unbiased estimate of the average effect of that accounting variable on future returns, and when the underlying variables are normally distributed, that unbiased estimate would characterize both the mean and median relation between y and x . However, since the distribution of returns is known to be skewed, the estimated OLS coefficient can be heavily influenced by a small number of extreme returns, which means that it could very well be the case that there is no relation between future returns and the accounting variable for the bulk of the observations. In this case, while OLS might be useful for predicting returns (i.e., developing a trading strategy), from a hypothesis-testing perspective, OLS can easily lead to an incorrect causal inference about the accounting variable due to the effect of the influential observations in the OLS regression.

From a model specification perspective, extreme values of the dependent variable caused by extreme values of the independent variable, called good leverage points, are not necessarily problematic because such y values are generated by the values of x . Where inference problems can arise are those cases where extreme values of y are not explained by x , something that can arise if such observations are the result of a different data generating process, where that different data generating process might be caused by an unknown or omitted economic variable that frequently takes on a value of zero, but which occasionally takes on a non-zero value, and when it does it has a major influence on y . In accounting, such observations are likely to occur frequently, and be the result of either one-time economic events or shifts in firm characteristics. To the extent that these economic events are correlated with the independent variables of interest, they lead to the

standard correlated-omitted variables problem, and depending on their magnitude, ignoring even low-frequency events can bias OLS coefficient estimates (hence inferences) as well as affecting standard errors.

The obvious challenge for researchers interested in testing hypotheses about the economic effects of accounting numbers is retaining those values of the dependent variable y (e.g., returns) that are “caused” by the independent variable of interest (e.g., earnings, etc.), while at the same time, limiting the influence of extreme values of y caused by infrequent economic events that are not included in the model. Retaining non-problematic leverage points is important from the perspective of maximizing power (i.e., ideally researchers should select designs that maximize good leverage points).¹⁰ In the next section we discuss how RR is designed to address these issues in ways that avoid ex-ante winsorization or truncation of the data.

4. Robust Regression (RR)

4.1 Properties of RR Estimators and Comparison with OLS¹¹

We refer to robust estimators, and particularly to RR, as a class of estimators that satisfy two conditions: “(1) if a small change is made to the data, it will not cause a substantial change in the estimate, and (2) the estimate is highly efficient under a wide range of circumstances” (see Andersen 2008, p. 3). The first condition for a robust estimator is its *resistance* to the presence of unusual observations. A resistant estimator provides a valid estimate for the bulk of the data. The second condition for a robust estimator is efficiency. An efficient estimator has high precision even when the underlying distributional assumptions of the estimator are not strictly met. An estimator is efficient if its variance is small, resulting in small standard errors. The literature on robust estimators has also focused on two additional properties, breakdown point and bounded influence. The breakdown point is an overall measure of the resistance of an estimator and is the smallest fraction of the data that a given estimator can tolerate without producing an inaccurate result. When an estimator “breaks down” it fails to represent the pattern in the bulk of the data. Bounded influence refers to the influence or marginal change of each individual observation on the properties of a given estimator.

¹⁰ As an example, when estimating earnings response coefficients, researchers should keep observations with large returns and large earnings surprises, which implies that it is not always desirable to winsorize or truncate large earnings surprises. Unfortunately, the winsorization and truncation rules used in accounting typically alter the distribution of earnings surprises without considering the value of y (returns in this case).

¹¹ Appendix A describes implementation of RR in more detail.

While widely used in accounting, OLS is not robust because its objective function is based on minimization of the sum of squares of the residuals, which increases indefinitely with the size of the residuals. By considering the sum of the squared residuals, OLS gives excessive importance to observations with very large residuals. In terms of the discussion above, OLS has unbounded influence. In fact, a *single* observation can have a significant impact on the fit of an OLS regression line/surface. This means the breakdown point of OLS is zero.¹²

An appealing feature of RR methods is that they estimate the parameters of a linear regression model while dealing with deviations from the OLS assumptions. While a number of RR techniques have evolved over time, our implementation is based on MM estimation. MM-estimators minimize the sum of a function of the scaled residuals using weighted least squares, where the weight function is non-decreasing for positive values and less increasing than the square function. What this means is that errors far from zero receive progressively less weight than errors closer to zero. The most commonly used weight functions are the Huber and bi-square functions. A key aspect of the weights given to each observation is that they are informative in the sense that they can be used to identify which observations are influential. MM-estimators are computed using an iterative procedure, because the residuals cannot be found until the model is fitted, and the parameter estimates cannot be found without the residuals. OLS is a special case within this class of estimators where the square function is used to weight the residuals. While MM-estimators are consistent, but less efficient than OLS, large sample efficiencies can be set close to 100% (use of 90 and 95% normal distribution efficiency is common in applied statistical packages like Stata).¹³

4.2 Issues related to implementation of RR

In this section we discuss issues related to RR implementation. While it is important to recognize these issues, we view them as issues that accounting researchers will be able to resolve in reasonably straight-forward ways as they become familiar with RR, as opposed to over-arching limitations preventing the implementation of RR in accounting. For example, there are a number of alternative RR estimation

¹² Consistent with this, some studies find that even a few observations can affect inferences based on OLS. For example, Guthrie et al. (2012) show that the results in Chhaochharia and Grinstein (2009) are driven by two observations in a sample of 865 firms, while Kraft et al. (2006) show that inferences about the underlying cause of the accrual anomaly are affected by roughly 200 observations (roughly 1% of the overall sample).

¹³ For proofs of the consistency and asymptotic normality of MM-estimators see Yohai (1987) and Maronna et al. (2006).

approaches (e.g., M-estimators, LP-estimators, and L-estimators). Fortunately, on this issue, accounting researchers can rely on the consensus in the RR literature that MM-estimation is preferred (see, Andersen 2008). Second, while RR implementation does not entirely eliminate the need for researcher judgment (in a sense, RR replaces the OLS minimization of squared residuals loss function for another loss function), with regard to researcher judgement, RR is arguably a more data-driven approach than are winsorization or truncation. Third, while the “canned” RR routines available in commonly used statistical packages do not automatically produce clustered standard errors, such standard errors are straightforward to generate using a bootstrap procedure like we use below in section 6. Fourth, RR is computationally more intensive than OLS, especially for models with a large number of variables and fixed effects. On this issue, we note that the cost of computing (speed of processors) continues to fall (increase). Finally, accounting researchers will need to become familiar with RR and there is a cost associated with that. Here we simply note that advances in econometric methods are a natural evolution of all research areas (e.g., accounting, Finance, economics, etc.) We provide additional guidance for implementation of RR in accounting in Appendix A.^{14,15}

5. Assessing the Effect of Influential Observations in Published Accounting Studies

We replicate three published papers to demonstrate how the choice of identification and treatment of influential observations can affect inferences. The studies are from different areas of the literature and analyze variables including return on assets, accruals, stock returns, and book-tax differences. Our objective is not to criticize or overturn these studies, but simply use their settings to illustrate how different approaches to identify and treat potential influential observations can affect coefficient estimates and inferences.

5.1 Replication #1: Accruals and earnings persistence

Richardson et al. (2005) (hereafter, RSST) examine the differential persistence of various accruals on earnings persistence, as well as the impact of such accruals on returns. We selected RSST because it uses variables that are important to researchers (earnings and accruals); it is widely accepted that the distributions

¹⁴ To facilitate implementing RR in accounting, the programs underlying all of our analysis are available upon request.

¹⁵ Relative to other econometric methods, we believe that RR is a more readily implementable than mixture models, latent class estimators, or even quantile regression. For example, while quantile regression is useful when the rate of change in the conditional quantile, expressed by the regression coefficients, depends on the quantile, quantile regression changes the quantity of interest and it not estimated using any type of least squares. Perhaps as a result of that, to date, it has not been widely used in accounting (although one application can be found in Armstrong et al. 2015).

of those variables contain extreme observations; and in their main analyses RSST use a different winsorization rule than the commonly used top and bottom 1%.

Our replication uses all COMPUSTAT observations with available data from 1988 to 2001, and as in RSST (Table 5, p. 461) we estimate (with all variables scaled by average total assets):

$$ROA_{i,t+1} = \beta_0 + \beta_1 ROA_{i,t} + \beta_2 \Delta WC_{i,t} + \beta_3 \Delta NCO_{i,t} + \beta_4 \Delta FIN_{i,t} + \varepsilon_{t+1}, \quad (2)$$

where for firm i and year t , ROA is operating income after depreciation; ΔWC is the change in net working capital (current operating assets - current operating liabilities); ΔNCO is the change in net non-current operating assets (noncurrent operating assets - noncurrent operating liabilities); and ΔFIN is the change in net financial assets (financial assets – financial liabilities). In Eq. (2) β_1 is expected to be positive and less than one because ROA is fairly persistent, while β_2 to β_4 are expected to be negative, with $\beta_2 < \beta_3 < \beta_4$, because RSST hypothesize that various accruals have different degrees of reliability. Descriptive statistics (available upon request) reveal that the variables of interest are all characterized by extreme observations and that winsorization and truncation result in distributions with different means and standard deviations (i.e., different properties) vis-à-vis the original raw data.

Table 2 reports coefficient estimates under alternative approaches to account for influential observations. Column (1) uses the original raw data (as it appears on the COMPUSTAT database available to us); column (2) uses data values that are winsorized at +1.0 and -1.0 for all variables (as in RSST, 2005 Table 5); column (3) uses data values winsorized at the 1st and 99th percentiles of each variable's distribution; column (4) uses data values where each variable has been truncated at its 1st and 99th percentile; column (5) uses values where the data is truncated based on a Cook's Distance (i.e., hereafter "Cook's D") greater than (4/Number of Observations). In Columns (1) to (5) estimation is based on OLS. In Column (6) the raw values of each variable (i.e., the original data) are used with RR based on MM-estimation.¹⁶

¹⁶ In columns (1) to (5) standard errors are clustered by firm. In Column (6) robust standard errors are estimated using a bootstrap-cluster procedure (1,000 replications) to cluster by firm. Clustering by firm is preferable, although qualitatively similar to the Fama-Macbeth approach used by RSST (p. 464) to mitigate the impact of auto-correlated errors over time (see, Gow et al. 2010). Under RR, we generate bootstrap standard errors by repeatedly sampling from the original data. Sampling was done by drawing firm-clusters with replacement to account for correlation between observations within a cluster where the bootstrap estimate of the standard error is the standard deviation of the bootstrapped sampling distribution. Estimating bootstrap-based clustered standard errors is consistent with the findings of Cameron et al. (2008) and the recommendations of Andersen (2008, p. 71). This approach can be extended to two-way (firm and year) clustering. See Appendix A for additional details (the code to generate boot-strapped standard errors under RR estimation is available upon request).

The estimates in Column (1) show that using the raw data yields a coefficient estimate that would seem to overestimate the persistence of ROA , since $\beta_1 > 1$. More importantly, the raw data shows no effect of accruals on earnings persistence (i.e., β_2 to β_4 are all insignificant). Column (2) shows the impact of winsorization at +1.0 and -1.0 (as in RSST) where we replicate RSST's results (i.e., β_2 to β_4 are all significant in the predicted direction). Column (3) shows the impact of the 1% and 99% winsorization rule (the cutoffs commonly applied in accounting) where we find that, β_2 , the coefficient on $\Delta WC_{i,t}$ (-0.126), is significantly negative at the 1% level (consistent with RSST); β_3 , the coefficient on $\Delta NCO_{i,t}$ (0.001), is positive, but not significant at the 10% level (inconsistent with RSST); while, β_4 , the coefficient on $\Delta FIN_{i,t}$ (-0.025), is significantly negative at the 1% level (consistent with RSST). Thus, under the winsorization approach commonly used in accounting, one would have drawn a different inference than RSST regarding the impact of $\Delta NCO_{i,t}$ accruals on earnings persistence. Column (4) shows the impact of the 1% and 99% truncation rule, the other approach commonly used in accounting. The results are similar to those in Column (2) in terms of the direction and statistical significance of the coefficients.¹⁷

The results reported in column (5) quantify the impact of using a regression influence diagnostic to identify influential observations, namely where observations with a “Cook’s D” larger than $(4/n)$, where n is the sample size) are excluded from the sample. The findings reveal that β_2 , the coefficient on $\Delta WC_{i,t}$ (0.031), is positive, but not statistically significant at the 10% level (inconsistent with RSST); β_3 , the coefficient on $\Delta NCO_{i,t}$ (0.116), is significantly positive at the 10% level (inconsistent with RSST); while, β_4 , the coefficient on $\Delta FIN_{i,t}$ (-0.060), is negative, but not statistically significant at the 10% level (inconsistent with RSST). Thus, if one had applied the Cook’s D influence diagnostic approach, he/she would have drawn different inferences than RSST regarding the impact of various types of accruals on earnings persistence.¹⁸ Finally,

¹⁷ The following subtle issue is worth noting. When truncation is based on the top and bottom 1% of *each variable's* distribution, roughly 7% of the total sample ends up being deleted. This means that researchers face the choice of deleting more observations as the number of models with different variables increases. While truncation yields similar results in this case, sample size is reduced by 7%, a reduction that may be unappealing in many accounting settings.

¹⁸ As an alternative to Cook’s D, we also used the $DFFITS > 2/\sqrt{(p/n)}$ criterion which yielded virtually identical coefficient estimates to those under Cook’s D. For example, β_2 , is positive (0.031), but not statistically significant at the 10% level (inconsistent with RSST); β_3 , is significantly positive (0.116) at the 10% level (inconsistent with RSST); while, β_4 , is negative (-0.060), not statistically significant at the 10% level (inconsistent with RSST). The similarity in results is not unexpected because Cook’s D and DFFITS are similar, except for a different scaling of the residuals. Moreover, the lowest value of both Cook’s D and DFFITS is zero and the higher the value of each, the more influential is a given observation.

without discussing them in detail, the results using RR (column 6) are similar to RSST, thus in this particular application of RR, estimation generally supports RSST's original inferences.¹⁹

In summary, while one might debate RSST's winsorizing the data at +1.0 and -1.0, because other accounting researchers' appear to have different priors about what constitutes acceptable cutoffs (based on the survey results reported in Table 1), such a debate is unnecessary and obscures the main takeaway of the replication. Namely, that the results demonstrate that RR yields coefficient estimates that are consistent with RSST's predictions without requiring an *ex-ante* approach to identify observations that may or may not actually turn out to be influential. That said, it is with noting that had RSST used the 1% and 99% winsorization rule commonly used in accounting, they would have reached a different conclusion about the differential effect of accruals persistence (i.e., they would not have concluded $\beta_2 < \beta_3 < \beta_4$).

5.2 Replication #2: Large book-tax differences and earnings persistence

Hanlon (2005) examined the impact of large book-tax differences on the persistence of pre-tax earnings. We selected Hanlon (2005) because it employs variables (e.g., book-tax differences and pre-tax earnings) whose distributions are characterized by extreme observations and because the study uses the 1% and 99% winsorization rule common to accounting studies.²⁰

Our replication uses all observations on COMPUSTAT with data available for the 1994 to 2000 period. Descriptive statistics (available upon request) reveal that the variables of interest are characterized by extreme observations. Following Hanlon (Table 3, p. 152) we estimate the following model:

$$PTBI_{i,t+1} = \beta_0 + \beta_1 LNBTD_{i,t} + \beta_2 LPBTD_{i,t} + \beta_3 PTBI_{i,t} + \beta_4 PTBI * LNBTD_{i,t} + \beta_5 PTBI * LPBTD_{i,t} + \varepsilon_{t+1}, \quad (3)$$

where for firm i in fiscal-year t , $PTBI$ is pre-tax book income (scaled by average assets); $LNBTD$ is an indicator variable equal to one for firms in the lowest quintile of book-tax difference (BTD) and zero

¹⁹ It is instructive to compare the coefficients across models to document whether there are any significant differences in coefficient magnitudes under the alternative approaches. To do this, for each model in columns (1) and (3)-(6) we tested whether the coefficient $WC_{i,t}$ (i.e., β_2) is significantly different from the estimate in column (2) of -0.094. We find the coefficients in columns (4) - (6) are all significantly different from the -0.094 value in column (2) (5% level). This result indicates that the choice of method also affects inferences via its impact on the magnitude of coefficient estimates.

²⁰ Another reason to select Hanlon (2005) is that Guenther (2011) identified 113 observations (2% of Hanlon's sample) with extreme book-tax differences and a large change in earnings, and when he removes those observations the earnings persistence of the remaining 98% of the sample is insignificantly different from that of small book-tax differences. That finding provides additional rationale for us to compare the alternative approaches in the book-tax difference earnings persistence setting.

otherwise (where *BTD* is deferred tax expense divided by the statutory rate of 35%, which is then scaled by average total assets); *LPBTD* is an indicator variable equal to one for firms in the top quintile of *BTD* and zero otherwise; and *PTBI*LNBTD* and *PTBI*LPTBI* are interaction terms. β_3 is expected to be positive and less than one given that earnings (scaled by average total assets) are fairly persistent, while the coefficients on the interaction terms, β_4 and β_5 , are expected to be negative for reasons hypothesized in Hanlon (2005).

Table 3 reports the results where column (1) uses the (original) raw data; column (2) winsorizes the data at the 1st and 99th percentiles of each variable's distribution (as in Hanlon 2005); column (3) truncates each variable at its 1st and 99th percentile; while column (4) truncates the data based on a Cook's D greater than 4/Number of Observations. Estimation in columns (1) - (4) uses OLS, while that in column (5) uses the (original) raw values of each variable with RR estimation.²¹

When the raw data are used (column 1), the interaction terms *PTBI*LNBTD_{i,t}* and *PTBI*LPTBI_{i,t}* are insignificant (inconsistent with Hanlon, 2005). When the data are winsorized (column 2) or truncated (column 3) both interaction terms are significantly negative at the 10% level or better. Under Cook's D (column 4), while the coefficient on *PTBI*LNBTD_{i,t}* (-0.079) is significantly negative at the 10% level, the coefficient on *PTBI*LPBTD_{i,t}* (0.091) is significantly positive (1% level). Under RR (column 5), only the coefficient on *PTBI*LNBTD_{i,t}* (-0.055) is significantly negative (1% level). The fragility of the results from model-to-model, notably, the coefficients on *PTBI*LPBTD_{i,t}* in columns (4) and (5) is evidence of influential observations affecting the inferences in Hanlon (2005) (see our footnote 20 as well as Guenther, 2011).^{22,23}

²¹ As before, in columns (1) to (4) standard errors are clustered by firm, while in column (5) robust standard errors are generated using the bootstrap procedure described above. As for our replication of RSST (2005), we also used the $DFFITS > 2/\sqrt{p/n}$ criterion instead of Cook's D which yielded virtually identical coefficient estimates to those using Cook's D.

²² Mechanically, interaction terms produce some collinearity between the main effects and the interaction terms. Since collinearity can produce changes in parameter estimates from sample to sample, or after minor data changes, we investigated whether collinearity might be driving the differences between the results for the winsorized data in column 2 and Cook's D in column 4. While the variance inflation factors (VIFs) for the model in column (2) range between 1.9 and 4.0 (indicating some degree of collinearity) such VIFs are not large when compared to the commonly used bound of 10.

²³ As before, it is instructive to compare the coefficients across models to see whether there are any differences based on the alternative approaches to account for influential observations. For each model in columns (1), and (3) - (5), we formally tested whether the coefficient on *PTBI*LPBTD_{i,t}* (β_5) is significantly different from the estimate in column (2) (where $\beta_5 = -0.066$). We find that the coefficients are statistically different (5% level) from the -0.066 in column (2) for columns (1), (4), and (5). These findings provide further evidence that choice of method to deal with influential observations also affects inferences via its impact on the magnitude of coefficient estimates, namely, in this case, whether β_4 and β_5 are negative.

5.3 Replication #3: Abnormal cash holdings and future performance

Oler and Picconi (2014) (hereafter, OP) study the effect of large positive and negative differences between actual and expected cash levels in a given year (i.e., excess vs. insufficient cash) on next year's profitability. OP are interested in whether extreme cash holdings have a negative effect on subsequent firm profitability. As before, we selected this study because it uses variables that are susceptible to extreme values and because OP use the 1% and 99% winsorization rule common to the literature. Another reason it is interesting to replicate OP is that they use a two-stage approach, namely, a "first-stage" model of cash holdings where fitted values are used to calculate "abnormal" cash holdings, which serve as an independent variable in a "second-stage" model of future profitability.

Our replication uses all observations from the intersection of CRSP and COMPUSTAT with data available for the 1989 to 2008 period. Following OP (Table 2, p. 12) we estimate the following model for the full sample and for rolling 5-year windows:

$$\begin{aligned} CASH_{i,t} = & \beta_0 + \beta_1 SIZE_{i,t} + \beta_2 NWC_{i,t} + \beta_3 CAPEX_{i,t} + \beta_4 INDSIGMA_{i,t} + \beta_5 RD_{i,t} + \beta_6 DIV_{i,t} \\ & + \beta_7 SALEGR_{i,t} + \beta_8 CFO_{i,t} + \beta_9 AGE_{i,t} + \beta_{10} TAXFI_{i,t} + \varepsilon_t \end{aligned} \quad (4)$$

where for firm i in year t , $CASH_{i,t}$ is cash and cash equivalents, scaled by total assets less cash; $SIZE$ is the natural log of book value of total assets; NWC is net working capital excluding cash, scaled by total assets less cash; $CAPEX$ is capital expenditures (from the Statement of Cash Flows), scaled by total assets less cash; $INDSIGMA$ is the average standard deviation of "cash flow" calculated for each 2-digit SIC code for the prior 20 years (since CFO is unavailable before 1988, pre-1988 CFO is defined as operating income before depreciation less interest, income taxes, and common dividends, scaled by total assets less cash, with missing values of interest, taxes, and dividends set to zero); RD is research and development expense scaled by total assets (missing values set to zero); DIV is an indicator variable equal to 1 if the firm paid common dividends over the prior year and zero otherwise (missing values set to zero); $SALEGR$ is current sales less prior year sales, scaled by prior year sales; CFO is cash flow from operations, scaled by total assets less cash; AGE is the natural log of the number of years the firm has been publicly traded measured as the difference between the firm's fiscal year less the date the firm first appeared on COMPUSTAT (plus one); and $TAXFI$ is foreign pretax income times the U.S. statutory rate of 35% less foreign income taxes paid multiplied by

100, then scaled by total assets less cash (which is set to zero if U.S. tax on foreign income is less than foreign taxes paid, or for missing values). Descriptive statistics (available upon request) indicate that all variables are characterized by extreme observations.

Table 4 reports the results of estimating Eq. (4) under the alternative approaches to account for influential observations. A key feature of the results is that the coefficients on $DIV_{i,t}$ and $SALEGR_{i,t}$ illustrate how inferences are affected depending on how influential observations are treated. For example, in contrast to column (2), where the coefficient on $DIV_{i,t}$ (-0.014) is insignificant, the models in columns (1) and (3) to (5) document a significant negative coefficient on $DIV_{i,t}$. In addition, in contrast to columns (2) and (5) where $SALEGR_{i,t}$ is insignificant, the model in column (3) shows a significantly positive coefficient of 0.016 (10% level), while that in column (4) shows a significantly negative coefficient of -0.001 (5% level).²⁴

Turning to the second-stage, as in OP, estimation uses rolling 5-year regressions of:

$$RNOA_{i,t+1} = \beta_0 + \beta_1 EXC_CASH_{i,t} + \beta_2 INS_CASH_{i,t} + \beta_3 NWC + \beta_4 SALEGR_{i,t} + \beta_5 LEV_{i,t} + \beta_6 SIZE_{i,t} \quad (5) \\ + \beta_7 RNOA_{i,t} + \varepsilon_{t+1},$$

where for firm i and year t , $RNOA$ is operating income after depreciation, divided by net operating assets where the industry mean $RNOA$ (2-digit SIC) is subtracted from the firm-level value; EXC_CASH is abnormal cash which is set to zero for the negative range of abnormal cash calculated as actual minus predicted cash (i.e., only positive and zero values remain); INS_CASH is abnormal cash which is set to zero for the positive range of abnormal cash (i.e., only negative and zero values remain); LEV is total debt scaled by total assets less cash (missing values set to zero); with all other variables as previously defined.^{25,26}

Table 5 reports the results of estimating Eq. (5) under alternative approaches to account for influential observations in both stages of the estimation. The results demonstrate that the approach used to account for influential observations impacts the association between abnormal cash levels and future profitability. Specifically (and in brief), using RR (see column 5) the coefficients on $EXC_CASH_{i,t}$, (-0.001)

²⁴ As for our RSST and Hanlon replications, DFFITS results in virtually identical coefficient estimates to those using Cook's D, a result which holds here in both the first and second-stage model estimation.

²⁵ The sample size drops from the first to second-stage estimation due to the requirement that $RNOA$ be available in year $t+1$, along with five years of data from $t-1$ to $t-5$ to estimate the cash holdings model fitted to year t data.

²⁶ Since cash holdings are bounded at zero, a Tobit regression could be used to estimate the cash model. Since it is not straight-forward to estimate RR for tobit models we use OLS, from which we obtain similar coefficients to those reported in Table 2 of OP.

and $INS_CASH_{i,t}$ (0.042) are insignificant, which contrasts with those in columns (2) to (4) where the coefficients on these two variables tend to be significantly negative (10% level or better).²⁷

5.4 Conclusions from the Replications

As noted above, the purpose of the replications is simply to illustrate how coefficient estimates and inferences can be affected by the approach a researcher uses to identify and treat influential observations. Two important takeaways from our replications are the following. First, the results of the studies do vary across (i.e., are not robust to) different approaches to identify and treat influential observations. Simply put, this means that a researcher's choice matters because coefficient estimates and inferences are not invariant across methods. Had different methods given the same results/inferences, one could have concluded that the approach used to identify and treat influential observations, while conceptually important may not turn out to be a critical research design choice in each and every setting. Second, and more generally, the findings of the replications indicate that the approaches commonly used in accounting (i.e., winsorization and truncation) may not provide robust inferences. A simpler way of stating this is that the results of some accounting studies may be specific to (i.e., driven by) the approach taken to identify and treat potential influential observations.

5.5 How can influential observations be analyzed under RR and what can be learned?

Under RR, a researcher can readily observe the weight given to each observation. This means that he/she can see which observations are getting low or zero weight, and then investigate them to find out what is causing them. In this section we illustrate this process by analyzing some of the observations RR identified as influential in each our replications. To do so, we selected 50 influential observations from each replication by identifying those observations with a weight equal to zero in the RR estimation. We then examined the firm's annual report to uncover whether there was an economic explanation for, or an economic event related to that observation. Such economic events are essentially correlated-omitted variables. Appendix B contains the results of this investigation where, for each replication, we report on the nature of any economic events related to each of the 50 observations. We briefly summarize our findings here.

²⁷ To be fair to OP, because they had concerns about extreme observations in their sample, we note that OP addressed that concern by also using decile ranked variables (scaling them from zero to one).

For our replication of RSST (2005), we found 40 cases where specific economic events would impact the variables of interest. As shown in Panel A of Appendix B we uncovered a number of events would have affected total assets (ROA denominator), including write-downs, M&A activities, and spinoffs. We also uncovered a number of events that would have affected net income (ROA numerator), including large one-time charges and write-downs, bankruptcy and reorganization, discontinued operations, large changes in R&D, and litigation-related events. For our replication of Hanlon (2005), we identified 48 cases where specific events impacted the book-to-tax difference variables of interest. As shown in Panel B of Appendix B, these included bankruptcy and reorganization, litigation, large changes in year-over-year income, and fifteen cases where an observation was from an IPO firm-year. Turning to Oler and Picconi (2014), we found 43 cases where specific events impacted cash holdings. For example, as shown in Panel C of Appendix B, 41 firms issued stock or obtained new financing. Given the prominence of new financing in these influential observations and the implications of new financing for interpreting OP's abnormal cash measures, to further investigate the nature of these influential observations we decided to gather additional information for five firms (primarily from their Statement of Cash Flows). The results are reported in Panel D of Appendix B. The common thread connecting all five is the inflow of cash from actual external financing events, which has the potential to affect measures of abnormal or excess cash. An implication of our investigation of the underlying economic aspects of these influential observations is that a "new financing" variable was not included in the original OP model that was used to estimate excess cash. As a consequence, the inferences of the original OP tests would likely be affected because excess cash in a given year driven by the new financing (i.e., the omitted variable) is likely to be correlated with future investments and performance (i.e., the dependent variable of interest in OP's second-stage model).

In summary, our investigation into the underlying economic aspects of the influential observations identified by our three replications illustrates that such observations were not data errors or random events. To the contrary, they consisted of legitimate data values resulting from actual economic events, observations that, in all likelihood, would have been inappropriately altered or transformed by the winsorization and truncation rules commonly applied in accounting research.

6. Simulation Analysis

6.1 Description of the simulation procedure

In this section we use simulations to compare the properties of the alternative methods to identify and treat influential observations. Simulated data with known properties enables us to identify the critical features of the following approaches: i) “doing nothing” about influential observations (i.e., use the data “as is”); ii) apply the winsorization and truncation approaches commonly used in accounting; iii) use a regression influence diagnostic such as Cook’s Distance (Cook’s D); and, iv) use RR estimation. Our objective is to provide systematic evidence on the relative efficacy and trade-offs associated with the various alternatives.

We begin by specifying a data-generating process for the dependent variable y consisting of a linear combination of independent variables of interest X plus a random error term e :

$$y = \alpha + X\beta + e, \quad (6)$$

where $X \sim N(0,1)$; $e \sim N(0,1)$; α is zero; and β are the slope coefficients relating the elements of X to y . We refer to the data (i.e., samples) generated using this process as the “*Clean Data*” because it does not contain any influential observations or random errors; which we will later introduce into different simulated datasets.

To model influential observations we add a variable $z \sim N(3,1)$ to the above data-generating process:

$$y = \alpha + X\beta + e + z. \quad (7)$$

In (7), the researcher does not observe z , which means he/she cannot control for z by, for example, adding a control variable. We simulate samples where the incidence of z occurs at random, as well as where some of the z observations are correlated with the values of x . We refer to the simulated data (i.e., samples) where the incidence of z occurs at random as the “*Data Containing Random Errors*.” To construct these samples, z is zero for 98% of the sample and occurs at random for the remaining 2%. These simulations model a real-world dataset containing a small number of random errors, which mimics, for example, a dataset that has random data entry errors for some observations.²⁸

We also simulate data (i.e., samples) where z is zero for 98% of the sample, but non-zero in the remaining 2% of the observations, where in that 2% of the sample, z reflects a real economic event that

²⁸ This dataset can be viewed as analogous to a real-world dataset in accounting where some observations are data entry errors, resulting, for example, from mistakes in the recording of financial statement data into COMPUSTAT where decimal points are inadvertently shifted or digits transposed, etc.

occurs more frequently for low values of an x variable of interest. We refer to these data (i.e., samples) as the “*Data Containing Correlated Economic Events*.” In these data (i.e., samples), z can be thought of as an actual economic event that occurs infrequently, but when it does it occurs it causes large values of y . The motivation to simulate and analyze this data is that the causes of influential observations in accounting are invariably non-random economic events (i.e., they are not simply data errors).²⁹

6.2 Graphical illustration of the effect of influential observations

We begin by providing graphical evidence illustrating the key features of the alternative approaches to identify and treat influential observations. For ease of presentation, the graphical evidence is based on univariate analysis (i.e., just one x variable in the underlying model). Figures 1 - 6 plot the simulated data, estimated regression line, and coefficient estimates under the alternative approaches. In each plot, the square dots are the simulated influential observations where z is nonzero (i.e., where a subset of the observations are affected by an extreme economic event) while the round dots are all of the other observations. All of the figures are based the “*Data Containing Correlated Economic Events*” (as described above) for samples of 2,000 simulated observations for a single x variable, $x \sim N(0,1)$, along with a simulated economic shock variable, $z \sim N(3,1)$, that occurs at random for one-fourth of the observations with x values under -1.5 (which means 2% of the observations are affected). We illustrate the effect of influential observations on coefficient estimates and inferences when there is an association between x and y (i.e., true $\beta = 0.8$) as well as when there is no association between x and y (true $\beta = 0$).

Figure 1 shows the estimated OLS regression line when the (original) raw data are used “as is” and where there is no treatment of influential observations. The OLS estimate of β is 0.661, which is significantly different (1% level) from the true β of 0.8 (i.e., the estimate is biased downward). Figure 2 shows the estimated OLS regression line when the true $\beta = 0$. The OLS estimate of β is -0.139, significantly different (1% level) from its true value of 0.0 (i.e., the estimate is biased downward). The key takeaway from Figures 1 and 2 is they illustrate that influential observations affecting as little as 2% of a sample can significantly

²⁹ For example, actual economic events that generate extreme observations in stock returns are likely to occur more frequently for smaller firms, firms making major operational changes, firms in the early (or late) phase of their life cycle, and/or firms characterized by new or novel business models.

affect OLS coefficient estimates. Moreover, in figure 2, where there is no association between x and y , the influential observations induce a spurious association between x and y .

Figure 3 shows the estimated OLS regression line (true $\beta = 0.8$) when both x and y have been winsorized at their 1st and 99th percentiles. In Figure 3 we can see that winsorization causes a number of observations to be clustered at the minimum and maximum values of each variable (see the inner bounds shown in the plots). After winsorization, the round dots in the figures represent the normally occurring observations while the square dots represent influential observations. The key aspect of the figure to appreciate is that winsorization leaves many of the extreme values of y , caused by z , in the sample (they appear within the winsorization bounds, but away from the trend line or the bulk of the data). Consistent with that, the OLS estimate of β of 0.660, is significantly different (1% level) from the true β of 0.8 (i.e., the estimate is biased downward). Thus, winsorization does little to mitigate the effect of truly influential observations. To see this point more concretely, note that the β estimate using the raw data “as is” with no treatment for influential observations (figure 1) is 0.661, which is virtually identical to the value of 0.660 under winsorization in figure 3.

Figure 4 shows the effect of truncating x and y at their 1st and 99th percentiles. The estimated β is 0.620, which is significantly different (1% level) from the true β of 0.8 (i.e., truncation biases the estimate downward). Thus, like winsorization, truncation does little to mitigate the effect of influential observations.

Figure 5 illustrates the effect of using the Cook’s Distance (Cook’s D) influence diagnostic. Using Cook’s D in this setting results in observations with absolute residuals greater than $(4/n = 4/2000)$ being removed from the sample. Doing so yields an estimated β of 0.737, which is significantly different (1% level) from the true β of 0.8 (i.e., the estimate is biased downward, although not by as much as under winsorization and truncation). Although the estimated coefficient of 0.737 is closer to the true β of 0.8 than are the estimates based on winsorization or truncation, it is still the case that using Cook’s D would affect inferences because the resulting coefficient estimate is still significantly different from the true value of 0.8.³⁰

³⁰ It is worth noting that Verardi and Croux (2009, p. 442) note that “Cook’s distances only manage to identify isolated outliers and are inappropriate when clusters of outliers exists, where one outlier can mask the presence of another.”

Finally, figure 6 illustrates the outcome of RR estimation where the estimated β of 0.771 is insignificantly different from true β of 0.8. Under RR, observations considered to be influential will be down-weighted. While the estimated β is generated using RR applied to all of the data, figure 6 only shows observations with weights greater than 0.5. The round dots represent the normally occurring observations while the square dots represent simulated influential observations. What is important to note is that, under RR estimation, extreme values of y caused by “true” extreme values of x (i.e., good leverage points) are retained with full weight, while influential observations receive a lower weight. In summary, across the alternatives illustrated in figures 1 to 6, RR produces results that best capture the true association between x and y in the data in terms of coefficient sign, magnitude, and bias.

6.3 Simulation analysis comparing the alternative approaches to identify and treat influential observations

To perform the simulations reported in this section, we expand the X vector to three variables (x_1 , x_2 , and x_3) where each is distributed $N(0,1)$. This choice reflects the multiple regression models characterizing accounting studies. As noted, simulations evaluate the following approaches to identify and treat influential observations: 1) use the original data “as is”, that is, “do nothing”; 2) winsorize all variables at their 1st and 99th percentiles, 3) winsorize only the X variables at their 1st and 99th percentiles; 4) truncate all variables at their 1st and 99th percentiles; 5) truncate only the X variables only at their 1st and 99th percentiles; 6) apply the Cook’s Distance influence diagnostic which means delete observations with residuals larger than $4/n$ (where n is the sample size; $n=10,000$ in our simulations); and 7) use RR on the original data. Model estimation in (1) - (6) is based on OLS while (7) uses RR based on MM-estimation.

The alternative methods are compared across six different simulated settings. The settings capture various combinations of the magnitude of the relation between vary x_i and y (e.g., $\beta_i = 0$ or $\beta_i = 0.8$) and whether the sample contains no influential observations (i.e., the “*Clean Data*”) or where the data contain influential observations (i.e., the “*Data Containing Correlated Economic Events*”). The six simulated settings are where: 1) no influential observations occur in the data and the true association between x_i and y is 0.8 (i.e., true $\beta_i = 0.8$); 2) no influential observations occur in the data and there is no association between x_i and y (i.e. true $\beta_i = 0.0$); 3) influential observations correlated with an economic event occur in the data and the true association between x_i and y is 0.8 (i.e., true $\beta_i = 0.8$); 4) influential observations correlated with

an economic event occur in the data and there is no association between x_i and y (i.e., true $\beta_i = 0.0$); 5) influential observations occur in the data, but do so *randomly* (i.e., are uncorrelated with an economic event), where the true association between x_i and y is 0.0 (i.e., true $\beta_i = 0.0$); and 6) influential observations occur in the data, but do so *randomly* (i.e., are uncorrelated with an economic event), where there is no association between x_i and y (i.e., true $\beta_i = 0$).

In samples where influential observations are correlated with an economic event, we introduce that influential observation by adding a variable $z \sim N(3,1)$ that occurs randomly for 25% of the observations with x_i values under -1.5 (which results in 2% of a sample's observations becoming influential). In samples where influential observations are present, but occur *randomly* (i.e., are uncorrelated with an economic event) we introduce those observations into the sample for 2% of the observations by randomly adding a variable $z \sim N(3,1)$. Each simulated setting is based on 1,000 independent samples of 10,000 observations each.³¹

For 1,000 samples of 10,000 observations each, the Panels of Table 6 report the mean coefficient estimate for β_i and the percent of the time across the 1,000 samples that mean coefficient is significantly different from its true value of 0.0 (or 0.80) based on 1% and 5% significance tests. These percentages correspond to the Type I error rate associated with each simulated setting and can be used to gauge whether the alternative approaches to identify and treat influential observations give rise to well-specified tests or whether they are biased against the null hypothesis that the estimated coefficient is equal to its true value. Evidence of a well-specified test would be a rejection rate at or near the 1% or 5% significance level of the test. Rejection rates above 1% or 5% (i.e., false positives) are evidence that a given approach is biased against the null hypothesis. Panel A reports results using the simulated "*Clean Data*", Panel B uses the simulated "*Data Containing Correlated Economic Events*", while Panel C uses the simulated data where influential observations are present, but occur randomly (i.e., are uncorrelated with an economic event).

Using the "*Clean Data*", columns (1) - (3) [(4) - (6)] of Panel A report results when the true $\beta_i = 0.8$ (0.0). In Column (1) all approaches, except winsorizing all the X variables and truncation of all variables, yield mean estimates of β_i that are close to the true value 0.8. Winsorizing all of the X variables imparts a

³¹ For comparison, Gow et al. (2010) simulated 1,000 datasets of 8,000 observations each. While accounting samples often contain more than 10,000 observations, it seems reasonable to presume that such studies do not have more than 10,000 independent observations (e.g., because they contain multiple observations on a given firm over time).

small upward bias (mean $\beta_I = 0.81$), while truncating all variables induces a pronounced bias, in this case a downward bias when the true coefficient is positive (mean $\beta_I = 0.75$). Turning to Type I errors, when the true $\beta_I = 0.8$, examination of columns (2) and (3) (5% and 1% significance tests, respectively) reveals that winsorizing only the X variables leads to poorly specified tests, with rejection rates of 26% (for a 5% test) and 11% (for a 1% test). Truncation of all variables does even worse, with rejection rates of 99% (for a 5% test) and 97% (for a 1% test). This is evidence that these alternatives to identify and treat influential observations should not be relied to draw inferences in accounting settings. With the exception of Cook's Distance, which exhibits rejection rates of 10% (for a 5% test) and 4% (for a 1% test), the remaining approaches exhibit rejection rates that closely correspond to the 5% and 1% significance levels of the test. Turning to Type I errors when the true $\beta_I = 0.0$ (columns [4] – [6]), Cook's Distance is the only mis-specified alternative exhibiting rejection rates of 10% (for a 5% test), and 4% (for a 1% test). In this setting, Cook's Distance is prone to induce a spurious association between x_I and y .

The simulation results reported in Panel B use the “*Data Containing Correlated Economic Events.*” Focusing on columns (1) - (3), which report results when the true $\beta_I = 0.8$, reveals that all approaches underestimate the true β_I of 0.8 (see column [1]). The approach that underestimates the true β_I of 0.80 the most (least) is truncation of all variables (RR) with a mean of 0.66 (0.77). Turning to rejection rates (columns [2] – [3]) all approaches exhibit rejection rates approaching 100%, except RR with 61% (for a 5% test) and 36% (for a 1% test). In columns (4) – (6), where the true $\beta_I = 0.0$, the results reveal that all approaches tend to result in a spurious association between x_I and y . The approaches with mean estimates of β_I that are the furthest from the true value of 0.0 are “do nothing” and winsorize only the X variables, both with mean estimates of -0.12. On the other hand, the approach with a mean estimate of β_I that is closest to the true value of 0.0 is RR (-0.03). To assess the significance of the mis-specified mean coefficient estimates in column (4), we turn to the Type I error rates in columns (5) and (6), where we find that all are excessive. Closer examination of the results reveals that RR exhibits the lowest (i.e., best) rejection rates (61% and 31%), while the rejection rates for all of the other alternatives approach 100%

Finally, Panel C of Table 6, reports results where influential observations occur, but do so randomly (i.e., are uncorrelated with an economic event). We discuss these results only briefly, simply because they are

very similar to the results reported in Panel A. For example, when the true $\beta_I = 0.80$, winsorizing only the X variables imparts a small upward bias (mean $\beta_I = 0.81$), while truncating all variables induces a pronounced downward bias (mean $\beta_I = 0.75$). In terms of Type I errors, when the true $\beta_I = 0.8$, winsorizing only the X variables leads to poorly specified tests (rejection rate of 25% for a 5% test and 9% for a 1% test). Truncation of all variables is even worse (rejection rates of 98% for a 5% test and 93% for a 1% test). This is further evidence that these alternatives should not be relied to draw inferences in accounting settings. Finally, as in Panel A, when the true $\beta_I = 0.0$ (see columns [4] – [6]), Cook's Distance is the only mis-specified alternative (rejection rates of 10% for a 5% test and 3% for a 1% test).³²

6.4 Results based on varying select simulation parameters

In this section we assess the sensitivity of earlier results by varying select simulation parameters. The parameters we vary are: 1) the magnitude of the relation between x_I and y (i.e., slope coefficient on x_I); 2) the magnitude of economic event, z ; and, 3) the percentage of observations in a sample experiencing the economic event, z . In the baseline simulations, the slope coefficient on x_I (i.e., true relation between x_I and y) was either 0.0 or 0.80. In Panel A of Table 7 we vary this from -0.80 to 0.80 in increments of 0.20. In the baseline simulations, we modeled the economic event, z as $\sim N(3,1)$. In Panel B of Table 7 we vary this from $N(1,1)$ to $N(5,1)$ (i.e., as $N[1,1]$, $N[2,1]$, $N[3,1]$, $N[4,1]$, and $N[5,1]$). Finally, in the baseline simulations, the percentage of observations experiencing the economic event, z , was set to 2% in each sample. In Panel C of Table 7 we vary this from 1% to 8%. All of these sensitivity tests use the “*Data Containing Correlated Economic Events*.” We briefly summarize the key findings of these sensitivity tests.

In Panel A of Table 7, the relation between x_I and y ranges from -0.80 to 0.80 (holding the percent of observations experiencing the effect of z at 2%, and the economic event, z , distributed as $N(3,1)$, both as in the baseline simulations). The most important feature of Panel A's results is that, regardless of the magnitude of β_I , the mean coefficient estimates based on RR are closest to the true value in all cases. The next best alternative is Cook's Distance, while the alternative that deviates most from the true value of β_I is the winsorize all x variables approach.

³² Results based on other influence diagnostics recommended by Belsley et al. (1980), namely DFFITS and DFBETA, are similar to those using Cook's Distance (available upon request).

In Panel B of Table 7, the magnitude (i.e., distribution) of the economic event, z , varies from $N(1,1)$, $N(2,1)$, $N(3,1)$, $N(4,1)$, through $N(5,1)$ (holding the true β_1 at 0.80, and the percent of observations experiencing the effect of z at the 2%, both as in the baseline simulations). The key findings are that: 1) as the magnitude of the event z increases (i.e., as we move from left to right across columns), the “doing nothing”, winsorization, and truncation approaches all produce mean estimates of β_1 that get further and further away from the true value of 0.8; 2) Cook’s Distance yields mean estimates that are effected by the extreme observations, but that effect is unrelated to the magnitude z since the mean estimate under Cook’s Distance is constant at 0.76 across columns (2) to (5); and perhaps most importantly, 3) that regardless of the magnitude of z , RR produces β_1 estimates that are the closest to the true of 0.8 (e.g., $\beta_1 = 0.77, 0.77, 0.79$, and 0.80 across columns 2 to 5).

Lastly, in Panel C, the percent of observations (in a given sample) experiencing the effect of z varies from 1% to 8% (holding true β_1 at 0.80, and the economic event, z , distributed as $N(3,1)$, both as in the baseline simulations). As in Panels A and B, Panel C’s main takeaway is that, regardless of the percentage of observations in a given sample that experience the event, z , mean coefficient estimates based on RR are closest to the true value of $\beta_1 = 0.80$ (with the next best alternative being Cook’s Distance).

6.5 Conclusions from the simulation analysis

The simulation results in Tables 6 and 7 lead us to four conclusions. First, truncation biases coefficients towards zero, and hence should be avoided. Second, winsorization does little to mitigate the effect of influential observations. Third, Cook’s Distance, while generally better than winsorization or truncation, is inferior to RR. Fourth, while none of the alternatives eliminates the effect of influential observations in each and every case, on average, RR performs the best in terms of mitigating coefficient bias and Type I errors. In summary, across a variety of simulated event conditions, our results indicate that RR is a viable alternative to address concerns about influential observations accounting research settings

7. Summary and Conclusions

We evaluated the efficacy and trade-offs associated with alternative methods to identify and treat influential observations in accounting research settings. We began by surveying the accounting literature and found the two most commonly used approaches are winsorization or truncation. Despite their widespread

use, there is little (if any) empirical evidence documenting whether they are effective at identifying and treating actual influential observations. In addition, our reading of widely-used econometrics texts (e.g., Greene 2006 and Wooldridge 2006) revealed that, neither one mentions (or advocates) variable-by-variable winsorization or truncation as recommended approaches to treat influential observations.

To document the efficacy of the following approaches to identify and treat influential observations -- winsorization, truncation, Cook's Distance regression influence diagnostic, robust regression (RR), and "do nothing" -- we used replications of published accounting studies along with simulation analysis. Our results documented that the choice of method affects the magnitude of coefficient estimates and their significance levels (hence statistical inferences based on them). On balance, the evidence indicates that RR generally outperforms the other approaches, especially in the presence of unusual (but actual) economic events that are correlated with the dependent and independent variables of interest. When compared to RR, truncation, winsorization, and Cook's Distance are not only much less effective in mitigating the effect of observations that are actually influential, but can also lead to bias in OLS regression coefficients.

While not widely used in accounting research to date, RR can be used in any situation where researchers would otherwise use OLS. In fact, we demonstrated that if there are no influential observations, RR and OLS produce similar coefficients and inferences. However, an added benefit of RR is that influential observations can be readily identified, hence subsequently studied to find out what might be causing them. This feature of RR is beneficial because it may lead to a revision in a theory being tested, or to development of a new or more robust theory that better explains the data.

In summary, while the evidence favors RR over the other approaches, we stress there is no panacea when it comes to identifying and controlling for influential observations in every conceivable accounting research setting. Nonetheless, our evidence should prove useful in the design of future accounting studies when it comes to making informed choices about how to address the presence of influential observations, and the consequences associated with choosing one alternative over another. Collectively, the findings of our replications and simulation analysis lead us to recommend that future accounting studies consider using RR, or at a minimum report sensitivity test results using RR.

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APPENDIX A

A Brief Description on Implementation of Robust Regression (RR)

1) General implementation guidance

This Appendix provides implementation guidance to facilitate use of RR in accounting research. To facilitate implementation of RR in accounting settings, the programs underlying all of our analysis are available upon request.

A common concern in accounting is cross-sectional dependence and its effect on standard errors. Currently available (i.e., “canned”) RR commands do not estimate clustered standard errors by default. However, as we did in this paper, one can estimate robust standard errors using a bootstrap method. Another alternative is using weighted OLS (based on the final RR weights) with clustered standard errors.

A variety of RR methods exist (see Andersen 2008 for a review) and within each method there are several estimation options. We used MM estimation, introduced by Yohai (1987), which combines high breakdown value estimation and M-estimation, which has both the high breakdown property and a higher statistical efficiency than S estimation. While, theoretically, there is a potential loss of efficiency using RR when compared to OLS, given that accounting studies typically have very large sample sizes, loss of efficiency is unlikely to be a concern when using RR in accounting.

In the presence of influential observations, adjusted R^2 is not necessarily the best criteria to assess model fit because it is based on squared deviations and influential observations usually have large deviations. Moreover, adjusted R^2 might be inflated when influential observations are leverage points (e.g., unusually large values of an x variable), while it might be deflated when influential observations are vertical outliers (e.g., unusually large values of y). As a result, as a general rule, it is not appropriate to compare the adjusted R^2 of RR with that of OLS. That said, in accounting the interest is generally in the sign and/or magnitude of the estimated coefficients (hence bias is important) as opposed to adjusted R^2 .

RR can be computationally intensive and models with a large number of indicator variables (fixed effects) may take time to estimate. In these cases, one could implement a modified RR algorithm that accommodates indicator variables (MS-estimation). Alternatively, researchers could start with a reduced RR model (without indicator variables), and next estimate a weighted OLS full model (with indicator variables) using the RR weights.

Logistic regression estimates can also be affected by influential observations, and while the primary focus of the RR literature has been regression models, there are estimators for when the dependent variable is binary (see Bianco and Yohai (1996) who propose a robust logistic regression estimator; see also Maronna et al. 2006, Chapter 7 for which Croux and Haesbroeck (2003) provide a computational algorithm).

2) Statistical packages that estimate RR

RR based on MM-estimation can be easily implemented, usually with just a few extra lines of code beyond OLS. For example, in STATA by using the `mm_regress` (Verardi and Croux 2009) and `robreg` (Jann 2010) commands (for documentation and examples see Verardi and Croux (2009) and Stata help). For models with dummy variables there is a variation of `mm_regress` called `ms_regress` (see the `mm_regress` documentation). In SAS, use the `proc robustreg` command (for documentation and examples see SAS Institute Paper 265-67). In R use the `robustbase` package and the command `lmrob` (see Rousseeuw et al. 2012) (documentation and examples are provided in the R software package).

There are two commonly used weight functions, Huber and Tukey bi-weight. Both the least-squares and Huber objective functions increase without bound as the residual e departs from 0, but the least-squares objective function increases more rapidly. In contrast, the bi-weight objective function eventually levels off (for $|e| > k$, where k is a constant). Least-squares assigns equal weight to each observation while the weights for the Huber estimator decline when $|e| > k$; and the weights for the bi-weight decline as soon as e departs from 0, and are 0 for $|e| > k$. The value k is called a tuning constant. Smaller values of k produce more resistance to extreme observations, but at the expense of lower efficiency when the errors are normally distributed. The tuning constant is generally picked to give reasonably high efficiency in the normal case; in particular, $k = 1.345\sigma$ for the Huber and $k = 4.685\sigma$ for the bi-weight (where σ is the standard deviation of the errors). The default function in the Stata “mmregress” command that we use is the Tukey bi-weight function with a tuning constant that achieves a breakdown point of 50% and efficiency of 95% (see Verardi and Croux 2009). Under these conditions, the weights look like an inverted “U” with flat parts at the beginning and end of the error distribution where only the most extreme influential observations are dropped (i.e., receive a weight of zero) and other influential points are down-weighted (i.e., receive weights less than one).”

The estimation procedure follows these steps: (1) first pass coefficients are calculated using some form of resistant regression (usually the S-estimator with Huber or bi-square weights); (2) the first pass coefficients are used to estimate residuals and the scale parameter; (3) a weight function is applied to the scaled residuals; (4) a second pass estimate of coefficients is obtained using weighted least squares; and (5) the new coefficients are used for a new iteration (keeping constant the measure of the scale of the residuals). The solution is considered to have converged when the change in estimates is no more than 0.1% from the previous iteration. MM-estimators are widely used and combine a high breakdown point with high efficiency.

3) Estimating clustered standard errors:

We have written a STATA ado file that computes two-way (firm and year) clustered bootstrapped standard errors for robgreg (computer programs are available from the authors upon request).

To estimate standard errors, it is possible to combine RR and the bootstrap with one-way clustering. For example, in STATA: `bootstrap "robgreg mm depvar xvar1 xvar2 xvar3" _b, cluster(firmid) reps(1000)`.

It is also possible (in STATA) to obtain the RR weights and estimate weighted OLS with clustering as follows:

```
robgreg mm depvar xvar1 xvar2 xvar3, generate(weight_rr)
reg depvar xvar1 xvar2 xvar3 [weight=weight_rr], robust cluster(firmid)
```

This latter alternative approach can be useful when a model has a large number of fixed effects (e.g., firm fixed effects).

4) Researcher specified settings:

Each RR algorithm has various settings that can be modified by the researcher. An important setting is the choice of efficiency rate. Efficiency refers to variation in estimated parameters from sample to sample. RR has up to 95% relative efficiency compared to the “best-case” scenario, that is, when OLS errors are normally distributed. There is a trade-off between the efficiency of a robust estimator when errors are normally distributed and the resistance of the robust estimator to influential observations. The above algorithms [see discussion above under (1)] have presets between 70% to 85% efficiency and a 50% breakdown point.

APPENDIX B

Economic characteristics of the influential observations identified from our replication of Richardson et al. 2005, (Hanlon, 2005), and (Oler and Picconi, 2014)

Panel A (Richardson et al. 2005), Panel B (Hanlon, 2005), and Panels C and D (Oler and Picconi, 2014) report the nature of the economic events underlying a sample of 50 influential observations from each of the three replicated studies. Influential observations are those firm-year observations with a weight equal to zero in the RR estimation. For each firm-year, annual reports were examined to uncover the existence of any economic events or economic explanations that might explain the influential observation.

Panel A: Accruals and earnings persistence (replication of Richardson et al. 2005) N=50

<i>Number of Instances</i>	<i>Description of Economic Event</i>	<i>Number of Instances</i>	<i>Description of Economic Event</i>
11	Asset write-down	2	Discontinued operations
7	Merger and acquisition activity	7	Large change in assets
1	Spin-off	1	Litigation
4	Bankruptcy and reorganization	1	Restatement
3	Change in research and development	2	One-time charges
1	Change in royalty revenue	10	Other/unclear event

Panel B: Large book-tax differences and earnings persistence (replication of Hanlon, 2005) N=50

<i>Number of Instances</i>	<i>Description of Economic Event</i>	<i>Number of Instances</i>	<i>Description of Economic Event</i>
9	Bankruptcy and reorganization	15	Initial public offering
11	Litigation	2	Other/unclear event
13	Large change in income		

Panel C: Modeling cash holdings (replication of Oler and Picconi, 2014) N=50

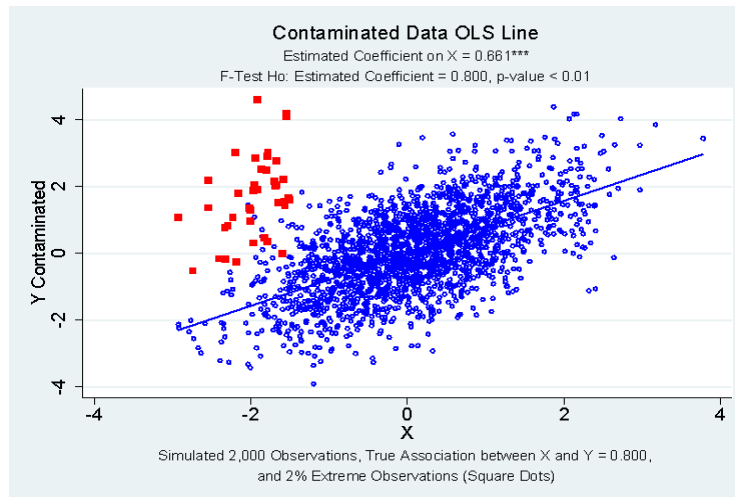
<i>Number of Instances</i>	<i>Description of Economic Event</i>	<i>Number of Instances</i>	<i>Description of Economic Event</i>
41	New financing	1	Merger and acquisition activity
1	Spin-off	7	Other/unclear event

Panel D: Detailed analysis of 5 influential observations from the replication of Oler and Picconi, 2014

<i>Firm Name</i>	<i>Total Assets (in \$M)</i>	<i>Cash and Equivalents (in \$M)</i>	<i>New financing (in \$M)</i>	<i>Brief description of the additional information uncovered on the effect of the firm's new financing on its cash holdings</i>
Exar Corp.	\$499.35	\$401.04	\$276.80	Issuance of 3,450,000 shares of common stock through a follow-on offering for net proceeds of \$260.8 million and \$16.0 million from the issuance of 1,337,036 common stock shares upon the exercise of stock options. Source: Statement of Cash Flows. (CIK 753568, Year 2000)
Xoma Corp.	\$64.78	\$55.15	\$22.10	\$10.0 million advance from Genentech under a long-term credit arrangement to fund 1998 development costs of \$10 and \$12.1 million. The company is attempting to build up cash while they pursue FDA licensure. (CIK 791908, Year 1997)
Printware Inc.	\$16.55	\$12.22	\$6.50	IPO in July of 1996. The company raised \$6.5 million from the issuance of its common stock. Source: Statement of Cash Flows. (CIK 801529, Year 1997)
Vical Inc.	\$50.69	\$45.55	\$4.40	Proceeds of \$4.4M from the issuance of common stock. Source: Statement of Cash Flows. (CIK 819050, Year 1997)
Vical Inc.	\$94.53	\$66.49	\$21.00	\$21.0 million in proceeds from the company's registered direct offering which closed in October 2005. Source: Statement of Cash Flows. (CIK 819050, Year 2005)

Figure 1
OLS Estimation with No Treatment for Influential Observations Using the “Data Containing Correlated Economic Events” when the True Association between x and y , $\beta=0.8$

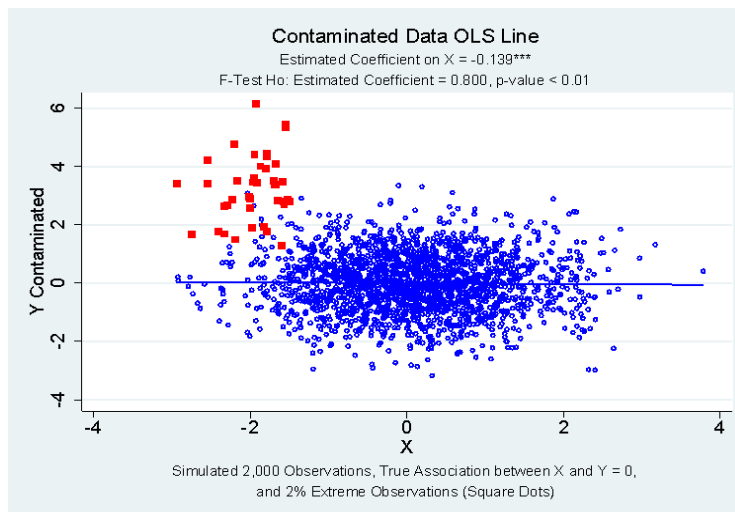
(The OLS estimate of β is 0.661, significantly different (1% level) from the true β of 0.8)



This figure illustrates how influential observations (i.e., “Data Containing Correlated Economic Events”) (the square dots) departs from the rest of the data (the round dots) when the economic events do not occur randomly, but instead are more frequent for small values of x . The figure is based on simulated data consisting of 2,000 values of y generated from the process: $y = bx + e$, where $x \sim N(0,1)$ and $e \sim N(0,1)$, where an “extreme economic event” z , and its effect on y , is simulated by generating an alternative value for y , from the process $y = bx + e + z$, randomly for one-fourth of the x values under -1.5 where $z \sim N(3,1)$. This results in extreme values in 2% of the y observations in the low end of the x distribution. The true β is set to 0.8 and an estimate of b is generated using the following OLS model: $y = \alpha + \beta x + e$.

Figure 2
OLS Estimation with No Treatment for Influential Observations Using the “Data Containing Correlated Economic Events” when the True Association between x and y , $\beta=0.0$

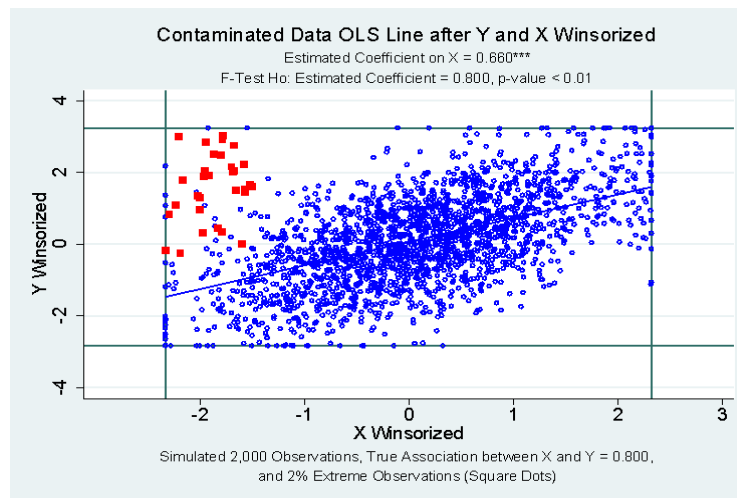
(The OLS estimate of β is -0.139, significantly different (1% level) from the true β of 0.0)



The data underlying the figure is generated in the same fashion as figure 1, except that the true β is set to 0. The figure illustrates how influential observations affect coefficient estimates even when there is no association between x and y .

Figure 3
OLS Estimation after Winsorization of x and y Using the “Data Containing Correlated Economic Events” when the True Association between x and y , $\beta = 0.8$

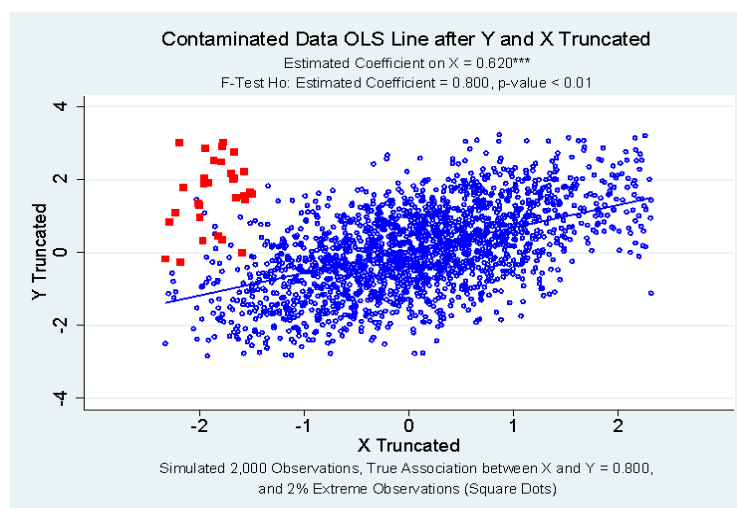
(The OLS estimate of β under winsorization of x and y is 0.660 significantly different (1% level) from the true β of 0.8)



The data underlying the figure is generated in the same fashion as prior figures. This figure illustrates how the data is transformed by winsorizing the x and y values at their 1st and 99th percentiles. As shown by the figure, winsorization causes a number of observations to be clustered at the minimum and maximum values of x and y . After winsorization, the round dots represent the normally occurring observations (no economic event) while the square dots represent simulated influential observations (an economic event).

Figure 4
OLS Estimation after Truncation of x and y Using the “Data Containing Correlated Economic Events” when the True Association between x and y , $\beta = 0.8$

(The OLS estimate of β under truncation of x and y is 0.620 significantly different (1% level) from the true β of 0.8)

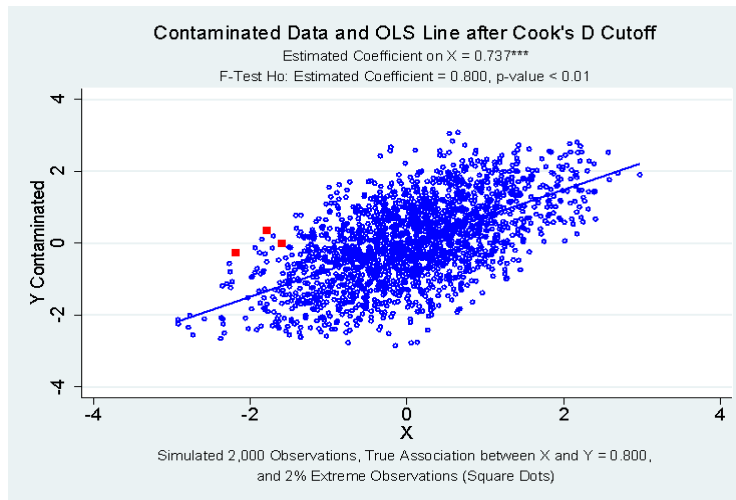


The data underlying the figure is generated in the same fashion as prior figures. The figure illustrates how the data is transformed by truncating the x and y values at their 1st and 99th percentiles. After truncation, the round dots represent the normally occurring observations (no economic event) while the square dots represent simulated influential observations (an economic event).

Figure 5

OLS Estimation after Applying the Cook's Distance Influence Diagnostic Using the "Data Containing Correlated Economic Events" when the True Association between x and y , $\beta = 0.8$

(The OLS estimate of β after applying Cook's Distance is 0.737 significantly different (1% level) from the true β of 0.8)

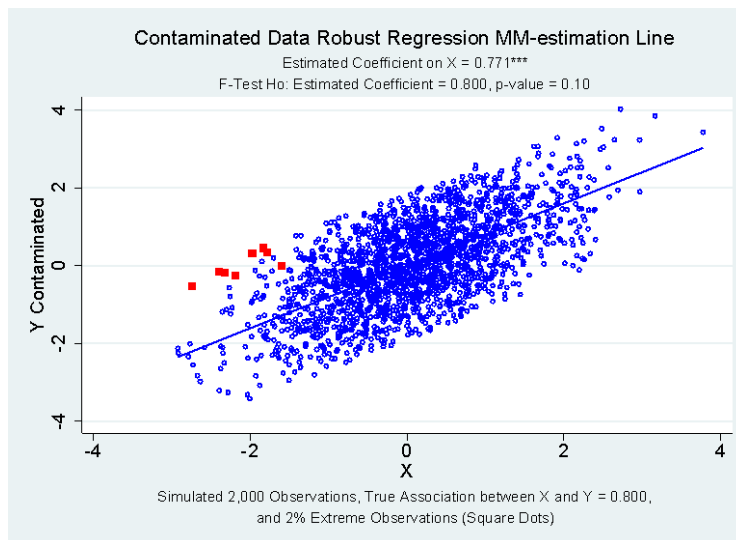


The data underlying the figure is generated in the same fashion as prior figures. The figure illustrates how the data is transformed when the Cook's Distance (Cook's D) influence diagnostic is applied. Using Cook's D, observations with absolute residuals greater than $(4/n)$ are dropped. After applying Cook's D, the round dots represent the normally occurring observations (no economic event) while the square dots represent simulated influential observations (an economic event).

Figure 6

Robust Regression Using the "Data Containing Correlated Economic Events" when the True Association between x and y , $\beta = 0.8$

(The RR estimate of β is 0.771, insignificantly different from the true β of 0.8)



The data underlying this figure is generated in the same as prior figures. The figure is based on using RR to identify and treat influential observations where an estimate of β is generated using RR applied to all of the data. The figure shows observations with weights greater than 0.5 where the round dots represent the normally occurring observations (no economic event) and the square dots represent simulated influential observations (an economic event).

Table 1

Survey Evidence on the Approaches Used to Identify and Treat Potential Influential Observations in Published Accounting Research *

Panel A: Studies by Research Paradigm

<i>Research paradigm:</i>	<i>N</i>	<i>% of Total</i>
Archival	1,252	68%
Analytical	161	9%
Experimental and survey	232	13%
Discussion and reviews	<u>188</u>	<u>10%</u>
<i>Total</i>	1,833	100%
<i>Archival studies:</i>		
Studies addressing or mentioning influential/extreme observations	851	68%
Studies not addressing or mentioning influential/extreme observations	<u>401</u>	<u>32%</u>
<i>Total archival studies</i>	1,252	100%
Studies using winsorization	532	63%
Studies using truncation	258	30%
Studies using both winsorization and truncation	59	7%
Studies using variable ranks	94	11%
Studies using other techniques (e.g., analysis of residuals and RR)	173	20%
(N and % are not additive because some studies use multiple approaches)		
Archival studies with returns as dependent variable	202	16%
Archival studies with dependent variables other than returns	<u>1,050</u>	<u>84%</u>
<i>Total archival studies</i>	1,252	100%
<i>Archival studies where returns is the dependent variable:</i>		
Studies winsorizing returns	46	23%
Studies truncating returns	46	23%
Studies using variable ranks of returns	18	9%
Studies using other techniques (e.g., analysis of residuals and RR)	29	14%
Studies winsorizing only the independent variables	37	18%
Studies truncating only the independent variables	9	4%
Studies using both winsorization and truncation	11	5%
Influential/extreme observations not mentioned	52	26%
(N and % are not additive because some studies use multiple approaches)		

* The table reports the findings of a survey for the discussion of and treatment of potential influential observations in studies published between 2006 and 2014 in the *Journal of Accounting Research*, *Journal of Accounting and Economics*, *Review of Accounting Studies*, *The Accounting Review*, and *Contemporary Accounting Research*. The studies reviewed span auditing, properties of analysts' forecasts, management compensation, earnings management, conservatism, tax, disclosure, and the earnings-return relation, etc. (studies with an analytical model and empirical tests are classified as archival).

Panel B: Characteristics of Archival Studies Using Winsorization

	<i>N</i>	<i>% of Total</i>
All variables (<i>x</i> and <i>y</i>)	337	63%
Dependent variable only (<i>y</i>)	32	6%
All independent variables (<i>x</i>)	51	10%
Leave out some variables (the stated winsorization rule is not to all variables of interest)	95	18%
Unclear cases (the text is not clear enough to determine which variables were winsorized, e.g., a study might state “we winsorize extreme values” without stating which variables)	<u>17</u>	<u>3%</u>
<i>Total archival studies using winsorization</i>	532	100%

Panel C: Characteristics of Archival Studies Using Truncation

	<i>N</i>	<i>% of Total</i>
All variables (<i>x</i> and <i>y</i>)	79	31%
Dependent variable only (<i>y</i>)	22	9%
All independent variables (<i>x</i>)	10	4%
Leave out some variables (the stated truncation rule is not to all variables of interest)	132	51%
Unclear cases (the text is not clear enough to determine which variables were truncated, e.g., a study might state “we truncate extreme values” without stating which variables that applied to)	<u>15</u>	<u>6%</u>
<i>Total archival studies using truncation</i>	258	100%

Panel D: Characteristics of a Specific Research Question/Issue: Research on Conservatism (N=57 studies)

	<i>N</i>	<i>% of Total</i>
Studies using winsorization	21	37%
Studies using truncation	20	35%
Studies using both winsorization and truncation	3	5%
Studies using variable ranks	9	16%
Studies using other techniques (e.g., analysis of residuals and RR)	8	14%
Influential/extreme observations not mentioned	17	30%
(N (%) does not sum to 57 (100%) because some studies use multiple approaches)		

Table 2
Accruals and Earnings Persistence Analysis (Replication of Richardson et al. 2005, Table 5)

		(1)	(2)	(3)	(4)	(5)	(6)
		<i>Dependent Variable = $ROA_{i,t+1}$</i>					
	<i>Predicted</i>	<i>COMPUSTAT</i>	<i>Winsorized</i>	<i>Winsorized</i>	<i>Truncated</i>	<i>Truncated</i>	<i>Robust</i>
<i>Variables</i>	<i>sign</i>	<i>(Raw Data)</i>	<i>at -1 and +1</i>	<i>at 1% and 99%</i>	<i>at 1% and 99%</i>	<i>Cook's D</i>	<i>Regression</i>
$ROA_{i,t}$	+	1.111	0.825***	0.841***	0.839***	0.866***	0.857***
		(1.11)	(198.3)	(140.9)	(137.9)	(14.22)	(146.8)
$\Delta WC_{i,t}$	-	-0.192	-0.094***	-0.126***	-0.129***	0.031	-0.069***
		(-0.73)	(-12.06)	(-10.21)	(-13.57)	(0.52)	(-9.14)
$\Delta NCO_{i,t}$	-	0.036	-0.028***	0.001	-0.030***	0.116***	-0.032***
		(0.22)	(-5.76)	(0.03)	(-5.61)	(2.96)	(-10.60)
$\Delta FIN_{i,t}$	-	0.170	-0.027***	-0.025***	-0.024***	-0.060	-0.018***
		(0.69)	(-7.33)	(-4.53)	(-5.09)	(-1.29)	(-4.38)
<i>Intercept</i>		-0.221***	-0.008***	-0.013***	-0.008***	-0.043***	0.010***
		(-2.82)	(-13.4)	(-16.62)	(-12.75)	(-8.53)	(18.40)
<i>Observations</i>		76,585	76,585	76,585	71,238	76,523	76,585
<i>Adjusted R²</i>		0.011	0.671	0.666	0.637	0.238	

The table reports coefficient estimates for Eq. (2) in the text using various approaches to account for influential observations. Column (1) uses the raw data for each variable as originally reported on COMPUSTAT; Column (2) winsorizes each variable at +1 and -1 (as is RSST, 2005); Column (3) winsorizes each variable at the 1st and 99th percentiles of its distribution; Column (4) truncates each variable at the 1st and 99th percentiles of its distribution; Column (5) uses the Cook's Distance influence diagnostic which truncates observations with a Cook's D greater than (4/N); and Column (6) uses the values of the raw data and robust regression (RR) based on MM-estimation. Estimation in Columns (1) to (5) is based on OLS with standard errors clustered by firm. Under RR estimation, standard errors clustered at the firm level based on a bootstrap procedure that uses 1,000 replications. Variable definitions are: $ROA_{i,t \text{ or } t+1}$ is Operating income after depreciation; $\Delta WC_{i,t}$ is the change in net working capital (current operating assets – current operating liabilities); $\Delta NCO_{i,t}$ is the change in net non-current operating assets (non-current operating assets – non-current operating liabilities); and $\Delta FIN_{i,t}$ is the change in net financial assets (financial assets – financial liabilities). All variables are scaled by average total assets. *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 3
Large Book-Tax Differences and Earnings Persistence Analysis (Replication of Hanlon 2005, Table 3)

		(1)	(2)	(3)	(4)	(5)
		<i>Dependent Variable = $PTBI_{i,t+1}$</i>				
	<i>Predicted</i>	<i>COMPUSTAT</i>	<i>Winsorized</i>	<i>Truncated</i>	<i>Truncated</i>	<i>Robust</i>
<i>Variables</i>	<i>Sign</i>	<i>(Raw Data)</i>	<i>at 1% and 99%</i>	<i>at 1% and 99%</i>	<i>Cook's D</i>	<i>Regression</i>
$LNBTD_{i,t}$	+	0.027 (1.23)	0.029*** (9.98)	0.029*** (10.24)	0.023*** (10.30)	0.019*** (8.86)
$LPBTD_{i,t}$	+	-0.070 (-1.39)	0.001 (0.27)	-0.001 (-0.44)	-0.018*** (-7.15)	-0.007*** (-2.84)
$PTBI_{i,t}$	+	0.666*** (3.82)	0.691*** (56.99)	0.689*** (58.68)	0.715*** (67.22)	0.830*** (80.39)
$PTBI*LNBTD_{i,t}$	-	-0.091 (-0.51)	-0.103*** (-4.97)	-0.104*** (-5.08)	-0.079*** (-5.01)	-0.055*** (-3.01)
$PTBI*LPBTD_{i,t}$	-	0.428 (1.16)	-0.066** (-2.38)	-0.043* (-1.68)	0.091*** (4.31)	-0.002 (-0.09)
<i>Intercept</i>		0.037* (1.75)	0.032*** (23.22)	0.032*** (24.29)	0.030*** (24.61)	0.015*** (15.34)
<i>Observations</i>		13,909	13,909	13,424	13,625	13,909
<i>Adjusted R²</i>		0.426	0.513	0.503	0.565	

The table reports coefficient estimates for Eq. (3) in the text using various approaches to account for influential observations. Column (1) uses the raw data for each variable as originally reported on COMPUSTAT; Column (2) winsorizes each variable at its 1st and 99th percentiles (as in Hanlon, 2005); Column (3) truncates each variable at its 1st and 99th percentiles; Column (4) uses the Cook's Distance influence diagnostic which truncates observations with a Cook's D greater than (4/N); and Column (5) uses the values of the raw data and robust regression (RR) based on MM-estimation. Estimation in Columns (1) to (4) is based on OLS with standard errors clustered by firm. Under RR estimation, standard errors clustered at the firm level based on a bootstrap procedure that uses 1,000 replications. Variable definitions are: $PTBI_{i,t \text{ or } t+1}$ is pre-tax book income; $LNBTD_{i,t}$ is an indicator variable equal to one for firms in the lowest quintile of book-tax difference (BTD) and zero otherwise, where BTD is deferred tax expense divided by the statutory rate of 35%; $LPBTD_{i,t}$ is an indicator variable equal to one for firms in the top quintile of book-tax difference (BTD) and zero otherwise; and, $PTBI*LNBTD_{i,t}$ and $PTBI*LPBTD_{i,t}$ are interaction terms. $PTBI_{i,t}$ and $BTD_{i,t}$ are scaled by average total assets. *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 4
Abnormal Cash Holdings and Future Performance Analysis (Replication of Oler and Picconi 2014, Table 2)

		(1)	(2)	(3)	(4)	(5)
		<i>Dependent Variable = CASH_{i,t}</i>				
	<i>Predicted</i>	<i>COMPUSTAT</i>	<i>Winsorized</i>	<i>Truncated</i>	<i>Truncated</i>	<i>Robust</i>
<i>Variables</i>	<i>sign</i>	<i>(Raw Data)</i>	<i>at 1% and 99%</i>	<i>at 1% and 99%</i>	<i>Cook's D</i>	<i>Regression</i>
<i>SIZE_{i,t}</i>	-	-0.134*** (-3.84)	-0.062*** (-14.62)	-0.040*** (-15.77)	-0.054*** (-15.74)	-0.006*** (-8.91)
<i>NWC_{i,t}</i>	-	-2.503*** (-3.07)	-0.920*** (-16.61)	-0.400*** (-15.35)	-0.815*** (-20.26)	-0.039*** (-7.12)
<i>CAPEX_{i,t}</i>	+	-1.385 (-0.83)	0.976*** (11.00)	0.749*** (11.92)	0.573*** (8.74)	-0.017 (-1.61)
<i>INDSIGMA_{i,t}</i>	+	0.400*** (4.79)	0.246*** (20.65)	0.169*** (19.06)	0.221*** (20.35)	0.021*** (11.65)
<i>RD_{i,t}</i>	+	-4.225** (-2.26)	1.302*** (11.73)	2.211*** (27.16)	0.559*** (4.47)	0.920*** (16.26)
<i>DIV_{i,t}</i>	-	-0.070* (-1.77)	-0.014 (-1.13)	-0.016* (-1.94)	-0.028** (-2.40)	-0.007*** (3.50)
<i>SALEGR_{i,t}</i>	+	-0.001 (-0.54)	0.014 (1.38)	0.016* (1.78)	-0.001** (-2.10)	-0.001 (0.01)
<i>CASHOP_{i,t}</i>	+	-1.549*** (-2.89)	-0.845*** (-22.76)	-0.504*** (-13.19)	-0.992*** (-19.71)	0.140*** (10.66)
<i>AGE_{i,t}</i>	-	-0.008 (-0.33)	0.022*** (3.11)	0.029*** (5.79)	0.024*** (4.63)	0.002** (2.42)
<i>TAXFI_{i,t}</i>	-	0.090** (2.26)	0.093*** (7.11)	0.025*** (2.71)	0.085*** (10.89)	0.028*** (8.99)
<i>Intercept</i>		1.662*** (3.02)	0.299*** (2.94)	0.009 (0.14)	0.230*** (4.26)	0.049*** (3.68)
<i>Observations</i>		64,391	64,391	55,408	63,704	64,391
<i>Adjusted R²</i>		0.378	0.438	0.291	0.428	

The table reports coefficient estimates for Eq. (4) in the text using various approaches to account for influential observations. Column (1) uses the raw data for each variable as originally reported on COMPUSTAT; Column (2) winsorizes each variable at its 1st and 99th percentiles (as in Oler and Picconi, 2014); Column (3) truncates each variable at its 1st and 99th percentiles; Column (4) uses the Cook's Distance influence diagnostic which truncates observations with a Cook's D greater

than $(4/N)$; and Column (5) uses the values of the raw data and robust regression (RR) based on MM-estimation. Estimation in Columns (1) to (4) is based on OLS with standard errors clustered by firm. Under RR estimation, standard errors clustered at the firm level based on a bootstrap procedure that uses 1,000 replications. Variable definitions are: $CASH_{i,t}$ is cash and cash equivalents scaled by total assets less cash equivalents; $SIZE_{i,t}$ is the natural log of book value of total assets; $NWC_{i,t}$ is net working capital excluding cash scaled by total assets less cash; $CAPEX_{i,t}$ is capital expenditures (as reported on the Statement of Cash Flows) scaled by total assets less cash; $INDSIGMA_{i,t}$ is the average standard deviation of “cash flow” calculated for each 2-digit SIC code for the prior 20 years (because CFO is not available until 1988, pre-1988 cash flow is defined as operating income before depreciation less interest, income taxes, and common dividends, scaled by total assets less cash where missing values for interest, taxes, and dividends are set to zero); $RD_{i,t}$ is research and development expense scaled by total assets (missing values set to zero); $DIV_{i,t}$ is an indicator variable equal to 1 if the firm paid common dividends over the prior year and zero otherwise (missing values set to zero); $SALEGR_{i,t}$ is current sales less prior year sales scaled by prior year sales; $CFO_{i,t}$ is cash flow from operations scaled by total assets less cash; $AGE_{i,t}$ is the natural log of the number of years the firm has been publicly traded measured as the difference between the firm’s fiscal year less the date that the firm was included in the COMPUSTAT file (plus one); and, $TAXFI_{i,t}$ is foreign pretax income times the U.S. Statutory rate of 35% less foreign income taxes paid multiplied by 100 and then scaled by total assets less cash (set to zero if the U.S. tax on foreign income is less than foreign taxes paid or for missing values). *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 5
Performance Regression Models: Abnormal Cash Holdings and Future Performance Analysis
(Replication of Oler and Picconi, 2014, Table 5)

		(1)	(2)	(3)	(4)	(5)
		<i>Dependent Variable = $RNOA_{i,t+1}$</i>				
	<i>Predicted</i>	<i>COMPUSTAT</i>	<i>Winsorized</i>	<i>Truncated</i>	<i>Truncated</i>	<i>Robust</i>
<i>Variables</i>	<i>sign</i>	<i>(Raw Data)</i>	<i>at 1% and 99%</i>	<i>at 1% and 99%</i>	<i>Cook's D</i>	<i>Regression</i>
<i>EXC_CASH_{i,t}</i>	-	-0.004	-0.024***	-0.027***	-0.008***	-0.001
		(-1.01)	(-7.45)	(-7.21)	(-2.60)	(-0.92)
<i>INS_CASH_{i,t}</i>	-	-0.035**	-0.012*	-0.087***	-0.070***	0.042*
		(-2.07)	(-1.84)	(-6.28)	(-4.76)	(1.77)
<i>NWC_{i,t}</i>	+	0.004	0.088***	0.025***	0.064***	-0.001
		(0.09)	(9.95)	(3.82)	(4.40)	(0.20)
<i>SALEGR_{i,t}</i>	-	0.001	-0.009***	0.011***	-0.001**	-0.001
		(0.71)	(-2.86)	(3.49)	(-2.22)	(0.84)
<i>LEV_{i,t}</i>	+	-0.047	-0.019**	-0.023***	-0.030***	-0.004
		(-1.40)	(-2.56)	(3.94)	(-3.44)	(1.15)
<i>SIZE_{i,t}</i>	+	0.032***	0.017***	0.010***	0.027***	0.002***
		(3.54)	(20.04)	(13.78)	(17.27)	(5.80)
<i>RNOA_{i,t}</i>	+	-0.016	0.479***	0.453***	0.143***	0.733***
		(-0.57)	(55.17)	(59.97)	(4.61)	(75.38)
<i>Intercept</i>		-0.129*	-0.090***	-0.033***	-0.109***	-0.005*
		(-2.46)	(-13.64)	(-6.21)	(-11.71)	(1.80)
<i>Observations</i>		43,009	43,009	38,906	42,850	43,009
<i>Adjusted R²</i>		0.002	0.293	0.237	0.317	

The table reports coefficient estimates for Eq. (5) in the text using various approaches to account for influential observations. Column (1) uses the raw data for each variable as originally reported on COMPUSTAT; Column (2) winsorizes each variable at its 1st and 99th percentiles (as in Oler and Picconi, 2014); Column (3) truncates each variable at its 1st and 99th percentiles; Column (4) uses the Cook's Distance influence diagnostic which truncates observations with a Cook's D greater than (4/N); and Column (5) uses the values of the raw data and robust regression (RR) based on MM-estimation. Estimation in Columns (1) to (4) is based on OLS with standard errors clustered by firm. Under RR estimation, standard errors clustered at the firm level based on a bootstrap procedure that uses 1,000 replications. Variable definitions are: $RNOA_{i,t \text{ or } t+1}$ is operating income after depreciation divided by net operating assets where the mean industry $RNOA$ (2-digit SIC) is subtracted from the firm-level value of $RNOA$; $AB_CASH_{i,t}$ is "insufficient/excess" cash calculated as actual cash minus the predicted level of cash; $INS_CASH_{i,t}$ is "insufficient/excess" cash set to zero for the positive range (which means only negative and zero values remain); $EXC_CASH_{i,t}$ is "insufficient/excess" cash set to zero for the negative range (which means that only positive and zero values remain); $NWC_{i,t}$ is net working capital excluding cash scaled by total assets less cash; $SALEGR_{i,t}$ is current sales less prior year sales, scaled by prior year sales; $LEV_{i,t}$ is total debt scaled by total assets less cash (missing values are set to zero); and $SIZE_{i,t}$ is the natural log of book value of total assets. *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 6
Results of Simulation Analysis Comparing Alternative Approaches to Identify and Treat Influential Observations

Panel A: Clean Data		True Association between x_1 and y ($\beta_1 = 0.8$)		No association between x_1 and y ($\beta_1 = 0$)		
		<i>Rejection $H_0: \beta_1 = 0.8$</i>			<i>Rejection $H_0: \beta_1 = 0$</i>	
<i>Alternative to account for influential observations:</i>	<i>Estimated β_1 Mean</i>	<i>p-value 5%</i>	<i>p-value 1%</i>	<i>Estimated β_1 Mean</i>	<i>p-value 5%</i>	<i>p-value 1%</i>
“Do nothing” (i.e., use data “as is”)	0.80	5%	1%	0.00	5%	1%
Winsorize all variables	0.80	5%	1%	0.00	6%	1%
Winsorize only the X variables	0.81	26%	11%	0.00	5%	1%
Truncate all variables	0.75	99%	97%	0.00	6%	1%
Truncate only the X variables	0.80	6%	1%	0.00	6%	1%
Cook's Distance influence diagnostic	0.80	10%	4%	0.00	10%	4%
Robust Regression	0.80	6%	1%	0.00	6%	1%

Panel B: Data Containing a Correlated Economic Event		True Association between x_1 and y ($\beta_1 = 0.8$)		No association between x_1 and y ($\beta_1 = 0$)		
		<i>Rejection $H_0: \beta_1 = 0.8$</i>			<i>Rejection $H_0: \beta_1 = 0$</i>	
<i>Alternative to account for influential observations:</i>	<i>Estimated β_1 Mean</i>	<i>p-value 5%</i>	<i>p-value 1%</i>	<i>Estimated β_1 Mean</i>	<i>p-value 5%</i>	<i>p-value 1%</i>
“Do nothing” (i.e., data “as is”)	0.68	100%	100%	-0.12	100%	100%
Winsorize all variables	0.68	100%	100%	-0.10	100%	100%
Winsorize only the X variables	0.69	100%	100%	-0.12	100%	100%
Truncate all variables	0.66	100%	100%	-0.05	98%	93%
Truncate only the X variables	0.69	100%	100%	-0.11	100%	100%
Cook's Distance influence diagnostic	0.76	98%	94%	-0.04	98%	94%
Robust Regression	0.77	61%	36%	-0.03	61%	36%

Panel C: Data where influential observations are present, but occur randomly (i.e., are uncorrelated with an economic event)		True Association between x_1 and y ($\beta_1 = 0.8$)		No association between x_1 and y ($\beta_1 = 0$)		
		<i>Rejection $H_0: \beta_1 = 0.8$</i>			<i>Rejection $H_0: \beta_1 = 0$</i>	
<i>Alternative to account for influential observations:</i>	<i>Estimated β_1 Mean</i>	<i>p-value 5%</i>	<i>p-value 1%</i>	<i>Estimated β_1 Mean</i>	<i>p-value 5%</i>	<i>p-value 1%</i>
“Do nothing” (i.e., use data “as is”)	0.80	6%	1%	0.00	6%	1%
Winsorize all variables	0.80	6%	1%	0.00	5%	1%
Winsorize only the X variables	0.81	25%	9%	0.00	6%	1%
Truncate all variables	0.75	98%	93%	0.00	6%	1%
Truncate only the X variables	0.80	5%	1%	0.00	5%	1%
Cook's Distance influence diagnostic	0.80	10%	3%	0.00	10%	3%
Robust Regression	0.80	6%	1%	0.00	6%	1%

The table reports results of comparing alternative methods of identifying and treating influential observations across six different simulated settings. The settings capture various combinations of the magnitude of the relation between x_1 and y (e.g., $\beta_1 = 0$ or $\beta_1 = 0.8$) and whether the sample contains no influential observations (i.e., “Clean Data”) or where the data contain influential observations (i.e., “Data Containing Correlated Economic Events”). The six simulated settings are: 1) no influential observations are present in the data and the true association between x_1 and y is 0.8; 2) no influential observations are present in the data and there is no association between x_1 and y ; 3) influential observations correlated with an economic event are present in the data and the true association between x_1 and y is 0.8; 4) influential observations correlated with an economic event are present in the data and there is no association between x_1 and y ; 5) influential observations are present in the data, but they occur *randomly* (i.e., are uncorrelated with an economic event), where the true association between x_1 and y is 0.0 (i.e., true $\beta_1 = 0.0$); and 6) influential observations are present in the data, but they occur *randomly* (i.e., are uncorrelated with an economic event), where there is no association between x_1 and y (i.e., true $\beta_1 = 0$). In those samples where influential observations are correlated with an economic event, we introduce that influential observation by adding a variable $z \sim N(3,1)$, that occurs randomly among 30% of the observations with x_1 values under -1.5 (which results in 2% of the sample observations becoming influential). In the samples where influential observations are present in the data, but occur *randomly* (i.e., are uncorrelated with an economic event) we introduce those observations into the sample for 2% of the observations by randomly adding a variable $z \sim N(3,1)$. The analysis of each simulated setting is based on 1,000 independent samples of 10,000 observations each.

Table 7

The Effect of Varying Select Simulation Parameters Using “Data Containing Correlated Economic Event”

Panel A: Varying the True Association between x_1 and y from -0.80 to 0.80 ($z \sim N(3,1)$ with 2% contamination)

Variation in the True β_1 coefficient:	$\beta_1 = -0.80$	$\beta_1 = -0.40$	$\beta_1 = -0.20$	$\beta_1 = 0.00$	$\beta_1 = 0.20$	$\beta_1 = 0.40$	$\beta_1 = 0.80$
Alternative to account for influential observations	Mean of Estimated β_1						
“Do nothing” (i.e., use the original data “as is”)	-0.92	-0.52	-0.32	-0.12	0.08	0.28	0.68
Winsorize all variables	-0.90	-0.50	-0.30	-0.10	0.09	0.29	0.68
Winsorize only the X variables	-0.94	-0.53	-0.33	-0.12	0.08	0.28	0.69
Truncate all variables	-0.82	-0.42	-0.23	-0.05	0.13	0.31	0.66
Truncate only the X variables	-0.91	-0.51	-0.31	-0.11	0.09	0.29	0.69
Cook's Distance influence diagnostic	-0.84	-0.44	-0.24	-0.04	0.16	0.36	0.76
Robust Regression	-0.83	-0.43	-0.23	-0.03	0.17	0.37	0.77

Panel B: Varying the Magnitude of the Event z , from $z \sim N(1,1)$ to $z \sim N(5,1)$ (true $\beta_1 = 0.80$ with 2% contamination)

Variation in the Distribution of event z :	$z \sim N(1,1)$	$z \sim N(2,1)$	$z \sim N(3,1)$	$z \sim N(4,1)$	$z \sim N(5,1)$
Alternative to account for influential observations	Mean of Estimated β_1				
“Do nothing” (i.e., use the original data “as is”)	0.76	0.72	0.68	0.64	0.60
Winsorize all variables	0.76	0.72	0.68	0.65	0.62
Winsorize only the X variables	0.77	0.73	0.69	0.65	0.61
Truncate all variables	0.71	0.68	0.66	0.66	0.67
Truncate only the X variables	0.76	0.73	0.69	0.65	0.61
Cook's Distance influence diagnostic	0.78	0.76	0.76	0.76	0.76
Robust Regression	0.78	0.77	0.77	0.79	0.80

Panel C: Varying the Percentage of Contaminated Observations from 1% to 8% ($z \sim N(3,1)$ and true $\beta_1 = 0.80$)

Variation in the percentage of contaminated observations:	1%	2%	3%	4%	5%	8%
Alternative to account for influential observations	Mean of Estimated β_1					
“Do nothing” (i.e., use the original data “as is”)	0.74	0.68	0.63	0.59	0.49	0.35
Winsorize all variables	0.74	0.68	0.63	0.59	0.50	0.36
Winsorize only the X variables	0.75	0.69	0.64	0.60	0.50	0.36
Truncate all variables	0.71	0.66	0.63	0.59	0.51	0.39
Truncate only the X variables	0.74	0.69	0.64	0.60	0.51	0.37
Cook's Distance influence diagnostic	0.78	0.76	0.74	0.71	0.66	0.55
Robust Regression	0.79	0.77	0.76	0.75	0.70	0.62

The table reports results from simulating 1,000 datasets with 10,000 observations each under alternative approaches to identify and treat influential observations. Use used samples *contaminated with correlated events* (referred to as the “Data Containing Correlated Economic Event”). We introduce influential observations or “contamination” by adding a variable z that occurs randomly among the observations with x values under -1.5. Panel A models the effect of variation in the association between x_1 and y , from -0.80 to 0.80, keeping $z \sim N(3,1)$ and 2% contamination. Panel B models the effect of variation in the magnitude of the event z , from $z \sim N(1,1)$ to $z \sim N(5,1)$, keeping $\beta_1 = 0.80$ and 2% contamination. Panel C models the effect of variation in the percentage of contaminated observations, from 1% to 8%, keeping $\beta_1 = 0.80$ and $z \sim N(3,1)$.